

CAPSTONE PROJECT 1
Proposal

**LexRAG-MY: Reducing Hallucination in Legal AI through Hybrid
Retrieval-Augmented Generation and Citation Verification for
Malaysian Legislation**

by

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Table of Contents

Abstract	7
Chapter 1: Introduction	9
1.1 Background Study	9
1.2 Problem Statement	10
1.3 Research Objectives	11
1.4 Research Questions	12
1.5 Scope and Limitations	12
1.5.1 Scope	12
1.5.2 Limitation	13
1.6 Significance of Research	13
1.6.1 Significance to Malaysian citizens	14
1.6.2 Significance to AI field	14
1.6.3 Significance to responsible and reliable AI practice	15
1.6.4 Significance to future researchers and students	15
1.6.5 Significance to Government and Institutional Deployment	15
Chapter 2: Literature Review	17
2.1 Introduction	17
2.2 Legal Question Answering and AI in Law	17
2.2.1 Evolution of AI in Legal Domain	17
2.2.2 Hallucination Risks	18
2.2.3 AI used in Malaysian Context	19
2.3 Retrieval-Augmented Generation (RAG)	19
2.3.1 RAG Architecture	19
2.3.2 Motivation of RAG	20
2.3.3 Evolution of RAG paradigms	21

2.4 RAG Applied to the Legal Domain	23
2.4.1 Case Law vs Statutory law RAG systems	23
2.4.2 Evidence of Retrieval and Faithfulness Failures	23
2.4.3 Solutions: Hybrid Retrieval and Citation Verification	24
2.5 Hybrid Retrieval Techniques (BM25 + Dense + Reranking).....	25
2.5.1 Sparse and Dense Retrieval and its Strength and Limitation	25
2.5.2 Hybrid Retrieval and Rank Fusion.....	26
2.5.3 Reranking Strength and Limitation	27
2.6 Citation Verification in RAG	29
2.6.1 Citation Recall and Precision	30
2.6.2 Citation Faithfulness.....	30
2.6.3 Verification Architectures	31
2.7 RAG for Non-English and Southeast Asian Contexts	32
2.7.1 Non-English Legal Context.....	32
2.7.2 Southeast Asian Legal Context	33
2.7.3 The Malaysian Gap	34
2.8 Comparison of Existing Work.....	34
2.9 Research Gap.....	37
Chapter 3: Methodology	40
3.1 Introduction	40
3.2 Overall System Architecture	41
3.3 Dataset Preparation	44
3.3.1 Source and Selection of Acts	44
3.3.2 PDF Parsing and Chunking.....	45
3.4 Retrieval Pipeline.....	46
3.4.1 Sparse Retrieval.....	47
3.4.2 Dense Retrieval.....	48

3.4.3 Hybrid Fusion (Reciprocal Rank Fusion).....	49
3.4.4 Reranking (Cross-Encoder).....	49
3.5 Answer Generation	50
3.5.1 Prompt Design and Context Constraint.....	50
3.5.2 Citation Generation	51
3.6 Citation Verification Mechanism.....	52
3.7 Evaluation Design.....	54
3.7.1 Test Set Construction	55
3.7.2 Retrieval Metrics (Precision@k and Recall@k).....	56
3.7.3 Generation Metrics (Faithfulness score using RAGAS)	57
3.7.4 Baseline Comparison	57
3.8 Tools and Technologies.....	58
3.9 Ethical and Practical Considerations	59
Chapter 4: Work Plan and Timeline.....	62
4.1 Introduction	62
4.2 Work Breakdown Structure (WBS)	62
4.2.1 Lists of Activities.....	63
4.2.2 Activities Dependencies and Critical Path	67
4.3 Project Milestones	70
4.4 Gantt Chart.....	71
4.5 Risk Considerations.....	72
Chapter 5: References	76
Chapter 6: Appendices	81
Appendix 1.0 The Evaluation Structure	81
Appendix 1.1 Test Set Structure.....	81
Appendix 1.2 Retrieval Score.....	81
Appendix 1.3 Generation Score	82

Appendix 1.4 Out of Scope Refusal Decision	82
Appendix 1.5 LLM Judge Validation	82
Appendix 1.6 Summary Table	83

List of Figures

Figure 2. 1 Overview of Retrieval-Augmented Generation	20
Figure 2. 2 Evolution of RAG Paradigms.....	22
Figure 2. 3 Hybrid Retrieval with RRF and Reranker	28
Figure 2. 4 Verification Gap for RAG system.....	31
Figure 3. 1 Full Architecture of LexRAG-MY	42
Figure 3. 2 User Mode and Batch Mode with a Shared Core Pipeline	43
Figure 3. 3 Example of Section Chunking Method using the Employment Act 1955	45
Figure 3. 4 Hybrid Retrieval Pipeline	47
Figure 3. 5 Citation Verification and Retry Flow	53
Figure 3. 6 Evaluation Design	55
Figure 4. 1 Work Breakdown Structure for LexRAG-MY	63
Figure 4. 2 Gantt Chart for LexRAG-MY	71
Figure 4. 3 Likelihood and Impact Matrix for LexRAG-MY	73
Figure A1. 1 Sample Structure of the Evaluation Test Set.....	81
Figure A1. 2 Retrieval Evaluation Template.....	81
Figure A1. 3 Generation and Citation Verification Template	82
Figure A1. 4 Out of Scope Evaluation Template	82
Figure A1. 5 LLM Judge Validation Template	82
Figure A1. 6 Summary Comparison of Retrieval Approaches Across All Evaluation Metrics.....	83

Abstract

Many Malaysians citizens including the employees, consumers and small business owners face legal questions, but they may not be able to afford professional lawyer consultation. Other alternatives may include searching through the official Malaysian statutory documents that is published publicly by the Attorney General's Chambers. Nevertheless, most laypeople do not understand the legal terminology from the documents. General purpose Large Language Models (LLMs) provides a cheaper solution, but they still hallucinate or provide a false answer based on assumption and produce a fluent yet inaccurate response that people have no way to verify. Hence, most people choose to remain silent, causing them failed to understand their own legal rights. Retrieval-Augmented Generation mitigates this by solving the high cost needed for consult a lawyer for minor issues, complex terminology when interpreting the official legal documents, and unfaithful response that produce by the LLMs. Yet, RAG does not guarantee zero hallucination rate. Additionally, existing research on RAG in legal domain were focused on United States case law and lack of research conducted on whether the RAG technique can be transferred to Malaysia statutory law, which the structure of case law is different from statutory law. This project proposed LexRAG-MY, a legal RAG system that is built on Malaysian statutory legal document, specifically the Employment Act 1955 and the Consumer Protection Act 1999 where both acts will be chunked at section level. Then, the retriever will combine both sparse and dense search to locate the most relevant chunks based on plain language, fused through Reciprocal Rank Fusion and further enhanced the result using a cross encoder reranker. Answer will then be generated only using the top-k chunks of results obtained in retrieval component using a constraint prompt, followed by a citation verification mechanism which the generated answer must pass through both existence check and support check before displaying the final answer to user. The proposed system aims to provide more people with access to accurate legal information. Hence, this system will be evaluated using 30 questions of self-built test set to ensure the reliability of system rather than assuming it. Metrics like Precision@k, Recall@k, RAGAS faithfulness score and LLM judge's agreement rate of proposed system will be compared across multiple simpler retrieval configurations. The study is expected to provide the first evidence on whether the RAG system with hybrid retrieval and citation verification component can improve answer reliability in Malaysian statutory context.

Keywords: Retrieval-Augmented Generation, Hallucination, Faithfulness, citation verification, hybrid retrieval, Malaysian statutory law, legal question answering

Chapter 1: Introduction

1.1 Background Study

Many Malaysians including the employees, consumers or even business owners often facing legal questions, but they are not able to consult a lawyer due to insufficient funds. It is not a small issue. As of 2025, number of employed persons in Malaysia's workforce continued to grow and reach over 17.09 million employed person (Department of Statistics Malaysia, 2026), but many employees still do not possess sufficient knowledge on legal rights awareness despite Malaysia is undergoing positive development of economic growth. A study in 2023 reveal that many workers have accepted salary reduction or unpaid leave, or even some employees gave up on their termination benefits during the COVID-19 period because there were unaware about their own rights and do not know that they were actually entitled for the benefits (Gan et al., 2023). Common Large Language Models like ChatGPT, Gemini, Claude or any Generative AI chatbot are the cheaper options and capable of answering questions across wide range of domains. Nevertheless, LLMs tend to hallucinate, provide answer based on assumption or produce outdated law and policy output which gained from trained memory (Yang et al., 2024), even with a legal AI tools that build specifically to answer legal question still archive hallucination rate of around 17% to 43% of the time (Magesh et al., 2025). These outputs appear to be convincing, but they are actually inaccurate. Alternatively, they can choose to check on official sources like Attorney General's Chambers (AGC) of Malaysia where all the Acts were published in it (Attorney General's Chambers of Malaysia, n.d.). However, this approach require user to be familiar with which sections within the Acts that is relevant to their situation while interpret and understand the complex legal terminology written in the Acts. This process requires large amounts of time and requires good understanding of legal language, which most laypeople do not possess. Retrieval-Augmented Generation (RAG) save the cost to consult a lawyer while producing a more trustworthy output from Generative AI (Magesh et al., 2025). It is an AI framework that retrieves information from an external database before generating outputs through LLMs (Y. Gao et al., 2023). RAG solves the hallucination problem that Generative AI chatbot usually encounters through retrieves relevant documents from database and only feeds the relevant information into LLM's prompt, while ensuring output is strictly generated based on the retrieved context (Lewis et al., 2020). The application of RAG

to legal industry has spark the interest in research lately due to the high accuracy in legal information. Several studies have explored the RAG based legal question answering (QA) system, but most of the work is concentrated on the United States (US) case law (Ho et al., 2025). QA refers to systems that answer questions. The research work of RAG based United States case law system may not work the same way as Malaysia statutory context as the strategy of fetching long structured narrative format may be different from methods designed to retrieve the exact paragraph in statute using the retrieval system (Li et al., 2023).

This means that the retrieval strategies between case law documents and statutory documents may not perform equally. This brings up the question on whether the RAG techniques that is largely develop on Case Law system can also be used in Malaysia Statute Law System to produce a more accurate and affordable answer to Malaysian citizens.

1.2 Problem Statement

Currently, Malaysian have three options when facing a legal question where each of them fails in different way. Consulting a lawyer is reliable but at the same time it might be costly for some direct legal question. Alternatively, laypeople can also choose to browse through the legal document manually from official sources, but it is time consuming to search the relevant documents based on their context. Furthermore, the complicated legal terminology written in the Acts may also discourage people to browse from the legal document. Using Generative AI to seek for legal advice could eliminate these issues but the answer may not be reliable because it may generate inaccurate information or providing outdated information, causing Malaysia citizen to misunderstand their own rights (Huang et al., 2025). To avoid troublesome, many of them choose to remain silent rather than seeking for a solution that they might actually have a valid right under a law. RAG could close the gap by enable LLMs to generate the answers using contents retrieved from documents instead of relying on the trained memory from Generative AI, which enhance the reliability and consistency of the output. Furthermore, RAG provides a more cost-effective solution as compared to hiring a lawyer, while reducing the time taken as opposed to searching for relevant information in formal documents. However, RAG does not guarantee to retrieve the

accurate information from those documents and may failed to locate the correct section based on query asked by user, especially if query asked is phrased in layman terms instead of legal terminology. Even if the correct sections are fetched into LLMs to generate output, there is no guarantee that the output produced by LLMs answered strictly based on the sections and may produce extra interpretation or assumptions (Noel et al., 2025). This means that RAG without a good architecture design still does not provide an accurate answer. Hence, to ensure the reliability of the system, RAG require a good retrieval strategy that handle layman terms, and a verification layer that ensure the output is traceable to the exact source retrieved.

These issues occur due to lack of research in Malaysia Statute Law System. Instead, most of the RAG-based legal questioning answering system research were developed and tested in United States with its Case Law system. Since the structure of the law is being written differently, the strategy of retrieving the exact paragraph in statute law using the retrieval system may be different from methods designed to fetch long structured narrative format of case law. As a result, Malaysian laypeople currently do not have a reliable, cost effective and a time efficient way to look for legal knowledge. Moreover, there is currently a lack of evidence on whether a RAG system with good retrieval and verification process can be reliable to answer Malaysia statutory law questions.

1.3 Research Objectives

- O1: To review existing RAG based system on legal domain literature and identify the technical gap on current system in terms of the reliability and quality of outputs and the research gap in applying RAG approach on Malaysia statutory law when most of the researched work on United States case law system that have different document structure as compared to statutory law.
- O2: To design a retrieval strategy that can locate the accurate parts of documents based on plain language legal questions.
- O3: To design a citation verification system that verifies the accuracy of the output of LLMs by checking the generated answer with the retrieved source section rather than produce extra interpretation or assumptions.

- O4: To evaluate the reliability of the system and check whether the answer generated is accurate based on retrieved section using test set of Malaysia statutory law questions and compare it with a simpler retrieval pipeline of RAG.

1.4 Research Questions

- RQ1: What are the limitations of existing RAG based system on legal domain and how far and what approaches does the RAG techniques have been applied in Malaysia statutory law system as compared to United States case law system?
- RQ2: How can the retrieval system be designed in a way that enable legal questions that are phrased on plain language be asked and locate the statutory sections accurately?
- RQ3: How can the citation validation mechanism be designed to ensure the answer generated by LLM is strictly based on the context extracted from the retrieval system only without making assumptions?
- RQ4: Does the proposed retrieval and citation validation system produce a more accurate and reliable answer as compared to simpler RAG retrieval pipeline when it is evaluated against test set of Malaysia statutory law questions?

1.5 Scope and Limitations

1.5.1 Scope

The project is limited to only two of the Malaysian Acts, which is the Employment Act 1955 and the Consumer Protection Act 1999 to ensure that the preparation of test questions can be used to validate the system within project timeline. It is developed as a web-based prototype with a chat interface that only allow the user to input their question in English. The system is designed to locate the relevant of sections within the Acts based on the question that may be phrased in plain or non-technical language. The answer generated from LLMs explains the retrieved section of Acts along with its citation, and the output are checked with the retrieved section again to confirm it is supported by the sources before user receives the answer. The project evaluates the system to ensure the accuracy and reliability of the answers though test set of

Malaysia statutory law questions and compare against a more simple retrieval approach to determine whether the proposed architecture is essential.

1.5.2 Limitation

The system will only be tested and answered based on the answers that clearly written within the two selected Acts. It cannot answer the questions with multiple interpretations that do not have a single objectively correct answer. It is only designed to answer based on the statutory sources provided and only limited to explaining what the relevant section states to provide a safer environment instead of allowing the system to reason about the user's personal context based on the question. Thus, it prevents mixing of information trained from LLMs, which leads back to the most crucial drawbacks of using Generative AI to search for outputs, hallucinations. Furthermore, it is not designed to replace professional legal advice for serious legal matters and do not include case law or any other types of law within the legal system. The system will not produce an output if the questions require legal reasoning based on the context. Lastly, it only supports English as input and only process each question independently without retaining context from previous question.

1.6 Significance of Research

The accessibility of seeking legal information has made it easier as people start to use Generative AI, yet this accessibility came with its consequences where these systems generate answer fluently that appear relevant to the question regardless of whether the answer is correct. This questioned the reliability of the output from Generative AI. Hence, in order to verify the accuracy of the output, layperson may choose to browse through the AGC of Malaysia that is time consuming. This research is significant to eliminate the gaps of unreliable answers using Generative AI and time consuming when manually read from official documents using RAG based system with retrieval strategy along with citation verification using Malaysia own statutory law as experimental platforms. The significance of the research can be view from four different perspectives, which is its significance to Malaysian citizens, its contribution to

Artificial Intelligence (AI) field, its relevance in responsible and reliable AI practice and its benefaction to future researchers and students.

1.6.1 Significance to Malaysian citizens

For most Malaysian citizens like employees, consumers or small business owners without a proper legal training, the method of seeking for legal information cannot be consulting a lawyer as questioned ask may not worth the cost of consulting a lawyer. Due to the constraints in cost and time, they might either chose to stay silence or seeking information through Generative AI. Both outcomes have their own disadvantages where people stay uninformed about their rights when keeping silence and might receive hallucinated outcomes when getting information from Generative AI. By returning outputs that is checkable citation to the source section, this system enable layperson to verify their answer without trusting solely on AI's confidence. This directly enhances the accessibility and reliability of true legal information without knowing any legal language and it eliminates the unnecessary cost for professional consultation for minor issues.

1.6.2 Significance to AI field

Retrieval-Augmented Generation is widely known to reduce hallucinations, but it still can retrieve from wrong section, do not retrieve anything or generate extra explanations based on what was retrieved (Huang et al., 2023). The existing research articles have implemented the RAG based QA system mostly on the case law from United States, which impose a question on whether the same technique can be transferred and used in Malaysia statutory law that has different structure and drafting style from case law. This research is important as it close that gap specifically in legal domain, as it is the first study to implement a RAG system with retrieval method and a citation verification method on Malaysian legislation. In addition, the system is used to compare with simpler retrieval pipeline of RAG using test set of Malaysia statutory law questions. Therefore, this research answer whether a good pipeline of RAG system is necessary and important by testing through measurable metrics rather than a general claim that RAG is beneficial.

1.6.3 Significance to responsible and reliable AI practice

Beyond its application to legal domain, this study also shows why retrieve and citation verification method using RAG is significant in other sensitive fields like medicine, law and engineering that might require reliable answers, where Generative AI may provide confident but hallucinate answers (Huang et al., 2025). The cost of error in specific domain is extremely high that will eventually cause injury, monetary or structural failure. This research does not assume RAG will always work. Instead, it builds a verification layer on top of the answer generated by LLMs to check the accuracy of cited section that truly supports the generated claim and measure it using test set of questions (Wallat et al., 2024). This contributes the evidence on the actual effectiveness as compared to simpler pipeline of RAG or even compare with answers generated by Generative AI. The technique may enhance the responsible and reliable of AI across various domains to receive an error free response.

1.6.4 Significance to future researchers and students

This research organises, clean and divided legal text of two Malaysian Acts (Employment Act 1955 and the Consumer Protection Act 1999) into chunks based on their sections, allowing future researchers ready to use in their research. This research also created a collection of tests set questions to evaluate the accuracy of the RAG system that is reusable to analyse throughout their analysis. These acts as a baseline for future students and researchers, allowing them to extend this work that is out of scope in this research. For example, future improvement may add case law or multilingual support in the system to increase the accessibility of this system.

1.6.5 Significance to Government and Institutional Deployment

Beyond individual citizens, LexRAG-MY could provide a meaningful implication on how government could provide legal information to citizen at scale. Currently, Malaysia government bodies like Jabatan Tenaga Kerja Semenanjung Malaysia or Tribunal Tuntutan Pengguna Malaysia receives large volume of inquiries about the rights under Employment Act 1955 and the Consumer Protection Act 1999 where most of the

questions were straightforward. Those questions do not require any human judgement and therefore several efforts like static FAQ pages and counter services have been made to provide legal information with minimal workload of government officers.

LexRAG-MY provides a more convenient alternative that could be deployed by government bodies on their official websites like AGC of Malaysia to provide citizens with an instant and accurate answers that is drawn directly from the official statutory text to reduce burden of routine question that need to be answered by human officers at a daily basis. As compared with commercial AI chatbots, this system ensures the transparency of output which the answer will be cited from the section of the Act. With that, it allows the response to be verified manually by citizens or government agencies.

Furthermore, some of the retrieval strategies used by commercial legal platforms like LexisNexis and Thomson Reuters Westlaw were not publicly disclosed, meaning that organizations that wish to adopt this technology cannot independently verified how the answers are generated (Magesh et al., 2025). LexRAG-MY, on the other hand, provides the full architecture approach, that is documented and made transparent in this research, making it replicable by any institution that wish to adapt or extend it. The evaluation in this project ensures the reliability of retuning a grounded answers for Malaysian statutory question. With a satisfactory result obtained during evaluation, it provides preliminary evidence that support future expansion towards deployment in official platforms, providing an accurate, accessible and intuitive response to citizens.

Chapter 2: Literature Review

2.1 Introduction

Chapter 1 has identified the problem that the efficiency of using RAG for Malaysian statutory law has remained unknown as prior research has focuses primarily on United States case law. This chapter focus on examine the relevant literature in order to support the claims while identified the choices for technical design and identify the research gaps.

The literature review begins with the use of AI in law and the use of QA system, then introduces the concept of Retrieval-Augmented Generation and how it is used in the legal domain. It then narrows down to reveal the existing techniques of RAG, including the hybrid retrieval and citation verification techniques. This chapter then compares the RAG used in legal domain among the non-western legal system, specifically the Southeast Asian, follow up by summarize the chapter that link towards the methodology and reveal the research gaps.

2.2 Legal Question Answering and AI in Law

2.2.1 Evolution of AI in Legal Domain

With the exponential rises of Artificial Intelligence technology over the past decades, its application has extended towards the law domain and evolve from rule-based system to data driven approaches and finally shift towards the generative approach recently. For example, Nguyen et al. (2024) developed an Attentive CNN and Paraformer mechanism to improve legal document retrieval performance while allowing questions to be asked in multiple languages like English, Japanese and Vietnamese. The issue then change from retrieval bottleneck to reliability bottleneck as AI develops and becoming more accessible. On the other hand, Lai et al. (2024) suggests that AI will eventually transform on how judicial industry works for the past few decades, but the application has remained at early development stage due to privacy infringement and lack of subjective opinion and emotions. The study pointed out that LLMs can now write answer fluently but might not be reliable which remain as unresolved problem.

Within the broad field of AI, legal QA system has its own study area. Development of Legal QA system is much more complex than general purpose QA system because it involves the complex terminology term used in legal domain (Kulkarni et al., 2026). Additionally, the answers generated must be traceable from the actual sources. Hence, the correctness of the answers alone is not enough but also requiring answers that can be traced back to origin sources. However, it is difficult to make comparison between systems since there is no consistent way to evaluate them. Guha et al. (2023) evaluate models using LegalBench benchmark that consist of 162 tasks that is standardized with LLM evaluation, specifying prompts, demonstrations, and metrics across six different legal reasoning types, but it can be only used in English. Louis et al. (2024) took a different method using Long-form Legal Question Answering (LLeQA) dataset to evaluate the system. It consists of 1868 statutory law question that is designed by experts but only limited in French language. The comparison between the two evaluation model have reflected that LegalBench with broad scope covers many types of legal reasoning, but it is too general to reflect on how an ordinary people could ask legal a question. Conversely, the LLeQA dataset was set based on official document with real legal questions, but it is too narrow as it only focuses on French and Belgian law. Both studies contributed and strengthened the evaluation technique in legal QA system, but neither of one cover the Malaysian legislation.

2.2.2 Hallucination Risks

Most researchers believe that retrieval method could fully eliminate the possibility of hallucination if AI searches from legal documents before generating outputs. Magesh et al. (2025) claims the statement are overstated by conducting an experiment with over 200 of legal questions and run the questions on Lexis AI and Westlaw AI which is a legal AI assistant, along with GPT-4 as general purpose Generative AI. Each of the three models had a hallucination rate of 0.17, 0.33 and 0.43. This result highlights that made up answers still occurred often on AI that used retrieval method as legal AI assistant even though it generally has lower hallucination rate than general purpose AI tools. Therefore, the result reveals that there might be chances of retrieving the wrong document or retrieving the right document but misunderstood it. As a results, the output that system generated fluently cannot simply be trust without checking

whether the answers are backed with the actual sites. Hence, it is significant and is not an optional to propose a citation verification system on top of AI retrieval method.

2.2.3 AI used in Malaysian Context

Taken together, these findings show that majority of legal AI research, which includes retrieval approach from Nguyen et al. (2024), and LegalBench and LLeQA evaluation methods were based in United States, Europe and East Asia, with lack of research covers the Malaysian statutory law. Legal QA system that performs well on other countries' law may not achieve similar results on Malaysia's different legal terminology and legal structure. On the other hand, research on AI in Malaysia legal system is limited and mainly used to assists on judicial applications instead of supporting ordinary people to gain access on legal information. For instance, similar study conducted by Khong and Ho (2022) described the use of AI system in court processes and criminal sentencing using machine learning algorithms that is developed by Sarawak Information Systems Sdn Bhd named Artificial Intelligence in Court Sentencing (AiCOS). This is in fact one of the limited research available for Malaysia AI legal system, with lack of research focus on information access for Malaysia statutory law.

2.3 Retrieval-Augmented Generation (RAG)

Section 2.2 demonstrated that answer from LLMs cannot be presumed based on the fluency of the answer as it might be hallucinated or making assumptions to generate a trustworthy answer even with tools that is designed for legal use. Thus, Retrieval-Augmented Generation (RAG) is used to address the problem, and it set the foundation architecture on this project is built.

2.3.1 RAG Architecture

RAG is a retrieved then generate architecture that is first introduced by Lewis et al. (2020) at Meta AI Research, that established a framework for modern enterprise AI. The paper categories into two types of knowledge, parametric memory and non-parametric memory. Parametric memory is the knowledge that is stored inside the

model's parameters where answers generated are based on the patterns it acquired during training. Non-parametric memory refers to external knowledge that is stored outside of the models. The basic architecture of RAG follows a simple loop as shown in Figure 2.1. It starts with receiving a query and convert it into a vector $q(x)$ using the query encoder. Then, it compares the vector $q(x)$ with the documents that is being stored as vector $d(z)$. Afterwards, it performs similarity search using Maximum Inner Product Search (MIPS) to determine the top-k relevant passages. Those passages were feed into LLMs to generate an answer based on the query and the retrieved context. Unlike standard Generative AI model, RAG combine both non-parametric memory to retrieved content from external documents source and parametric memory used to understand context and language before generating an answer. Their finding proves that this approach have produced more factual output than a parametric-only baseline.

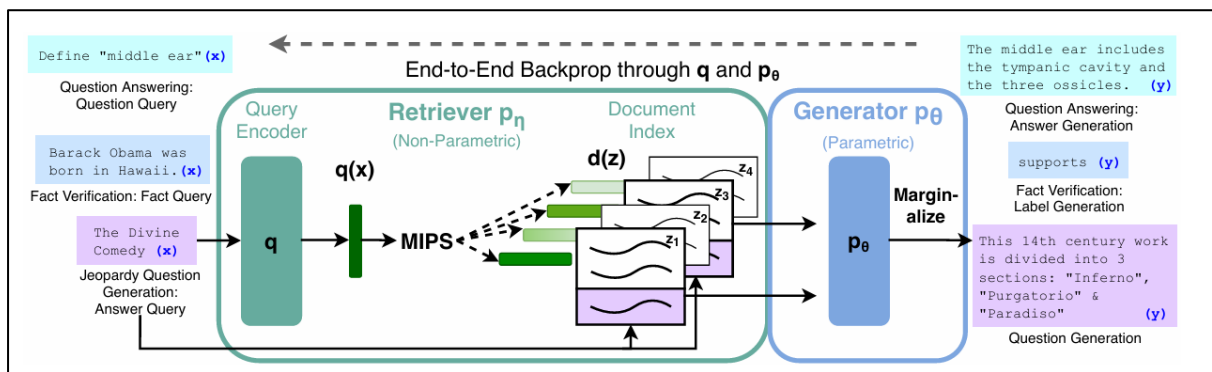


Figure 2. 1 Overview of Retrieval-Augmented Generation

2.3.2 Motivation of RAG

The main reason of RAG being developed is to reduce the hallucination problem that LLMs poses. Huang et al. (2023) portrays hallucination in AI as an output that appears fluent, but the actual information is misleading or inconsistent based on the real-world facts or its source due to outdated information or lack of understanding of the context. Hallucination is dangerous in legal or other sensitive applications as user need a precise information rather than a general interpretation. RAG reduce hallucination by providing the models with external information sources so that responses generated are based on evidence rather than only with its pre-trained knowledge. Conversely, RAG is not the only way to reduce hallucinate. Methods like prompt tuning, decoding

strategy and self-refinement through feedback and reasoning that were outlined by Tonmoy et al. (2024) can also be used to mitigate hallucinations. Yet, the characteristic of statutory law makes RAG the most suitable solution for the current issue due to the statute structure that is hierarchical and able to organize into parts and sections. Given the fixed identifier obtains from each part and sections, it is easier to split the whole content of documents into smaller chunks and stored separately. It is beneficial to RAG system when fetching a specific section to use it as evidence. Alternative methods like prompt tuning and decoding adjustment strategies only concentrating on fine-tuning the way it selects among the trained patterns instead of focusing on retrieving from external knowledge. The answer to statutory questions must be specific and traceable back to the formal document rather than being uncertain (Alshehri et al., 2026). This explains why RAG approach is used as the core architecture of this project rather than prompt tuning, changing the decoding strategy or self-refinement using existing models.

2.3.3 Evolution of RAG paradigms

To organize the approaches covered further in this chapter, Y. Gao et al. (2023) and Singh et al. (2025) examines the progression of RAG paradigms with three stages of development, as shown in Figure 2.2. Naïve RAG is a basic retrieved then generate architecture that focusing on keyword-based retrieval techniques like TF-IDF and BM25. While it is easy to implement, it failed to capture the semantic nuances of question. This issue can be resolved using Advanced RAG with its ability to improve retrieval quality through hybrid search for keyword and semantic retrieval. Besides, Advanced RAG also include post retrieval processing by using reranking and context pruning process to discard the irrelevant text. Modular RAG is a more sophisticated framework that is able to breakdown traditional pipeline into independent parts, ensuring the flexibility to swap component without rewriting the entire application. The proposed system in this project aligns with the classification of RAG paradigms, showing that improvement in architecture like incorporating hybrid retrieval and citation verification elements could contribute to higher and more consistent output quality which reduces the hallucination rates.

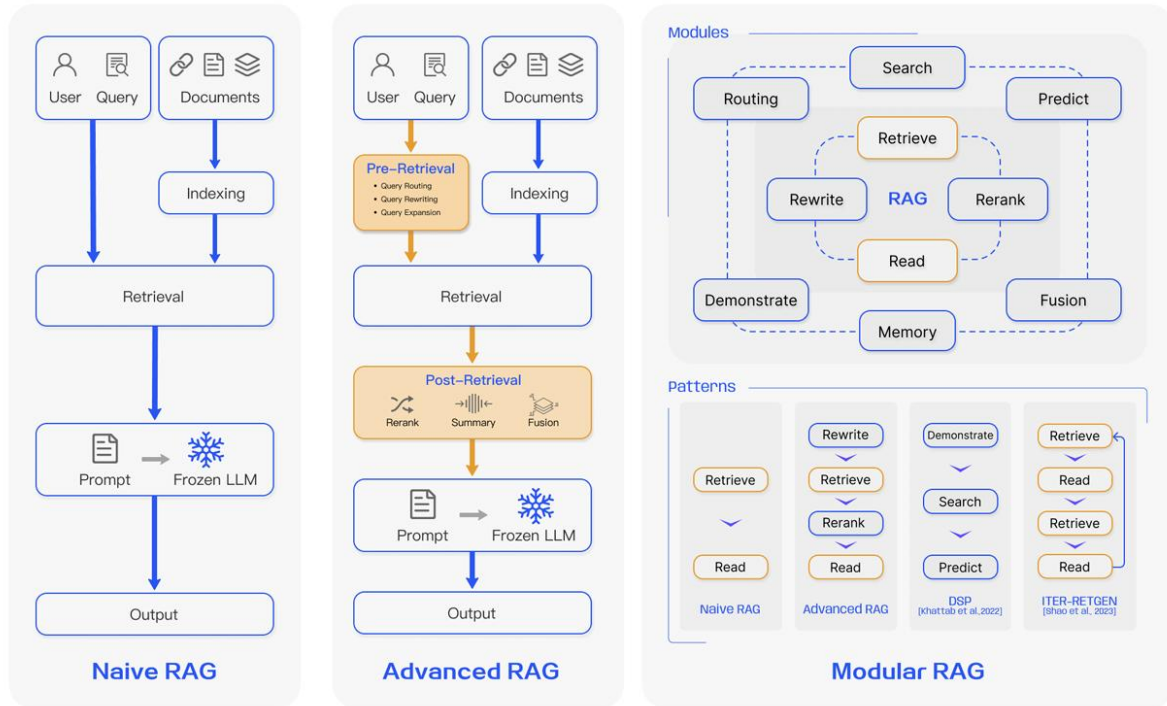


Figure 2. 2 Evolution of RAG Paradigms

While RAG improve the accuracy of retrieving relevant information, it does not guarantee a faithful and hallucination-free answer. Huang et al. (2025) and Tonmoy et al. (2024) explained the two points of failure, retrieval failure and faithfulness failure. While retrieval failure happens when correct information is not retrieved, resulted in incorrect context even before passing into LLMs, faithfulness failure refers to model introduce extra assumptions even when the relevant context is available. Because the basic RAG still has weaknesses, the Self-Reflective Retrieval-Augmented Generation (SELF-RAG), a framework proposed by Asai et al. (2023) is designed to improve the effectiveness of RAG by reflects its own output and retrieval quality before displaying the answer to user. The effectiveness of RAG consistently depends on the quality of the retrieved context, and the faithfulness of the generated answer. Together, these two factors were taken into account and become the inspiration for the system design proposed in this project.

2.4 RAG Applied to the Legal Domain

2.4.1 Case Law vs Statutory law RAG systems

While Section 2.3 concluded that RAG is an architecture that have two different points of failure, this section focuses on the real-world application of RAG used in the legal domain while comparing its performance in terms of their effectiveness and limitations. The retrieval method in legal domain should be divided based on different type of law. The separation is essential as different type of law have different challenges on retrieval. Most research done focused on case law, with Hou et al. (2025) as an example launched the CLERC dataset that consist of millions of cases that is used to evaluate RAG on case law. Fewer research focuses on statutory law, including the LLeQA dataset developed by Louis et al. (2024) focused on French and Belgian statutory law and LexRAG introduced by Li et al. (2025) with over 17 thousand articles from Chinese statutes. As mentioned in Chapter 1, there is a difference between long narrative structure of case law and short separate sections in statutory law. The difference in structures explains why results between each law cannot directly apply to each other.

2.4.2 Evidence of Retrieval and Faithfulness Failures

One way to determine the behaviour of RAG is to evaluate the legal RAG system that have been tested in real life. For instance, the experiment conducted on legal RAG assistant by Magesh et al. (2025) and RAG dataset launched by Hou et al. (2025) that built on US case law, which both are RAG that apply directly in legal domain. Section 2.2 suggested that tools with retrieval method might still hallucinate while Section 2.3 indicated the two hallucinate point as retrieval failure and faithfulness failure. These studies provide proof of these problems. CLERC proves the difficulty of retrieval, with BM25 retrieves 48.3 percent of the correct documents based on Recall@1000. This highlights that out of all the questions that have been tested, the correct documents were found in top 1000 of the system were only 48.3 percent of all the time. Even the best performing model do not fully eliminate hallucinations despite producing answers that is very similar to answers obtained from the actual documents. The legal RAG assistant executed by Magesh et al. (2025) shows the other side of failure where answer can still be unfaithful even though the correct sections are retrieved. Even in

real legal applications, RAG does reduce but not totally abolish the hallucination, so the system needs an extra verification step to ensure that the final answer is supported and able to be traced back from original sources.

2.4.3 Solutions: Hybrid Retrieval and Citation Verification

To solve this problem, newer system tries different retrieval approaches made for legal domain. Wiratunga et al. (2024) introduce the Case-Based Reasoning (CBR) RAG on Australian QA dataset and found out that hybrid retrieval strategy offers a more effective response as compared to single retrieval strategy. This encourages the use of hybrid retrieval pipeline in this project even though the procedure is conducted only using Australian legal documents and was not checked with multiple countries, so it cannot be expected that Malaysian statutory law will work the same way. For evaluation, Park et al. (2025) proposed LRAGE, a tool for RAG system specifically in legal domain. Their results reveal that rerankers that built for general RAG system may not make legal-based RAG better as it is trained in general domain which may not understand the legal text. This means that a reranker should not be added in this project solely because it performs better in normal QA system, but it should be evaluated before implementing it in legal RAG system.

Further research focus on whether the answers generated by LLMs cite the correct law contexts within the document, also known as the faithfulness of the output. LexRAG is the closest example to this project because it is tested on the Chinese statutory law by measuring the faithfulness of the answer rather than determining simply on whether the response sounds right. This validates that citing the correct section can be measured which apply the similar concept in the project with the citation verification approach.

Together with all the studies combined reveal its research gaps. Every system that was built, including CLERC on United States case law, CBR-RAG from Australia, LLeQA focuses on French and Belgian statutory law and LexRAG developed from Chinese statutes do not automatically apply in Malaysia statutory law context. LLeQA and LexRAG both uses statutory law in their system, which proves that statutory law in RAG is achievable. Nevertheless, lack of research covers the Malaysian law, and neither of articles combine both hybrid retrieval together with a citation verification

architecture on any RAG used in legal domains. This leads to the next sections that addresses the hybrid retrieval methods in Section 2.5 and the citation-verification methods in Section 2.6.

2.5 Hybrid Retrieval Techniques (BM25 + Dense + Reranking)

Section 2.4 established that the real word legal domain RAG system does encounter retrieval failure as one of the two points that makes RAG hallucinate which the correct section often overlooks even by big system. Therefore, the quality of retrieved context is important to contribute to an accurate answer because if the relevant sections never feed into LLMs, the final output will only be limited by the retrieved context regardless of how carefully the prompt is designed. No RAG system could produce output that are completely free from hallucination, but a good retrieval strategy could significantly lower the rate of hallucination (Wiratunga et al., 2024). With that, this section proposes the retrieval strategies available and compare a single method retrieval method with hybrid retrieval method, while also introducing techniques like rank fusion and reranking to enhance the overall model performance. Y. Gao et al. (2023) and Singh et al. (2025) categorized the combination of hybrid search and post retrieval reranking as the “Advanced RAG” that describe it as an evolution beyond the traditional RAG, also known as the “Keyword-based naïve RAG”.

2.5.1 Sparse and Dense Retrieval and its Strength and Limitation

Single retrieval method for either keyword or semantic retrieval failed to maximise their ability to capture the exact statutory sections. Keyword retrieval, also known as sparse retrieval, rank the exact words that occur in both query and document. BM25 is a retrieval algorithm that is widely used for keyword retrieval, with its strength of locating the exact section of the documents based on the exact legal terms or identifier such as “Section 32(1)”. However, when a layperson describes its situation in plain language rather than formal term used, the algorithm could not match the exact keyword, causing the correct section may not be retrieved at all (Singh et al., 2025). Semantic retrieval, or dense retrieval address the weakness by encoding both query and document into a shared embedding space and search for the specific section based on the similarity of meaning rather than exact wording (Singh et al., 2025). The

dual-encoder dense retrieval is more efficient than BM25 system by 9 to 19 percent in terms of top 20 passage retrieval accuracy using a variety of open-domain QA datasets, proving that dense retrieval outperforms the sparse retrieval method by capturing the semantic meaning that keyword-match search often misses (Karpukhin et al., 2020). Meanwhile, the study also highlights that BM25 still perform better on queries that rely on rare phrases, including the section numbers and legal terminology that is necessary for the answer.

2.5.2 Hybrid Retrieval and Rank Fusion

This creates a conflict particularly for this project as system are expected to received input in plain language, which aligned well using semantic retrieval. In contrast, the document that injected into RAG system consist of statutory sections and legal terminology which is better suited to keyword retrieval. Neither method used alone could match the context for a statutory legal RAG system that understand in simple language. This is not a question of which retrieval method perform better, as each method works well in different situations. Hence, the relationship between the two methods is complementary rather than mutually exclusive alternatives. The complementary strengths motivate the hybrid retrieval approach which run with both method and merge them into single results with Sawarkar et al. (2024) proved that the hybrid strategy outperforms either method used alone. This research proposed Blended RAG method that combine dense retrieval, sparse encoder indexes and BM25 using Natural Questions and TREC-COVID datasets, with the results demonstrated that the proposed method perform better than system designed for the targeted tasks. Mala et al. (2025) took a similar approach on combining sparse and dense retrieval methods and fuses the results through a weighted Reciprocal Rank Fusion (RRF) score to reduce hallucinations which enhances the reliability of generated answers. Wiratunga et al. (2024) reach a consistent conclusion that hybrid method improves the answer quality than single retrieval method on Australian legal documents. These studies reached an identical conclusion where hybrid retrieval has achieved better results than single method retrieval technique across both open and legal domain. Despite this, the evidence should be considered with caution as the CBR-RAG developed by Wiratunga et al. (2024) was not tested in other countries'

legal system while the Blended RAG method introduced by Sawarkar et al. (2024) was only tested on general domain datasets rather than legal documents. Therefore, the benefits of using hybrid retrieval methods may not directly apply to Malaysian statutory law.

After each of retriever return a ranked list, those lists were combined into single set of results. Reciprocal Rank Fusion, introduced by Cormack et al. (2009) combined the document relevancy ranking from multiple information retrieval system and gives a higher score for the documents that appear consistently on top of any list of sources. This led to overall better retrieval quality compared to any individual retrieval systems or earlier merging methods like Condorcet fusion. Additionally, it does not require any training data as RRF simply uses a mathematical rule to rank documents. This is why it is considered the most suitable option in Malaysian statutory legal setting, especially when there is lack of existing labelled question section training dataset that can be used to train a model.

2.5.3 Reranking Strength and Limitation

While fusion produces a strong set of results, it can be further sharpened by a reranking stage. Nogueira and Cho (2019) proposed the first BERT cross-encoder reranker that evaluate the relevance of section preselected by BM25, then encodes the question and section together as an input. This approach improves the ranking performance on MS MARCO, a benchmark for testing retrieval and reranking models. This paper established wide used of “retrieve then rerank” approach where sections can now be retrieved and rerank it uses a more accurate model. However, the cross-encoder reranker is costly even though it enhances the accuracy of sections. This is because it processes every question-section pair to produces a relevance score for each pair it is given. Hence, it encourages the reranker component to be implement on the candidates chunks rather than the whole corpus, which is exactly why it is placed after BM25 retrieval rather than using it as a primary retriever. Improved technique by Khattab and Zaharia (2020) offer an alternative way to rerank the relevancy of passages using ColBERT, which is an advanced version of BERT cross-encoder reranker. It encodes the documents into multiple precomputed token level

embedding, while query is encoded at input time. ColBERT uses a “late interaction” architecture where the query token is matched with all documents token, and the best matching score are combined to produce a final relevance score. This design avoids the repeated costs to run every query–document pair, which is a cheaper and more powerful approach. In terms of legal domain, Faisal et al. (2024) uses the “retrieve then rerank” technique that pair a BM25 with BERT reranker on Indonesian statutory regulations, reporting a 20 percent improvement in exact match accuracy. This paper uses reranker approach in Southeast Asia statutory law shows that it can be also apply in Malaysian statutory law. But the research uses only sparse retrieval as method to extract passage, which does not prove that hybrid retrieval would also work well on this kind of statutory text.

However, the efficiency of reranker should not be presume as Park et al. (2025) reported that reranker that trained on general domain retriever may reduce the accuracy of RAG system and misinterpret the legal relevance. It should not be incorporated in RAG solely because it improves the performance on open domain context, but it should be validated on Malaysian statutory evaluation process before drawing conclusion about the efficiency of the reranker, which emphasizes the significance of comparing a full pipeline with a simpler retrieval pipeline as part of an objective in this project.

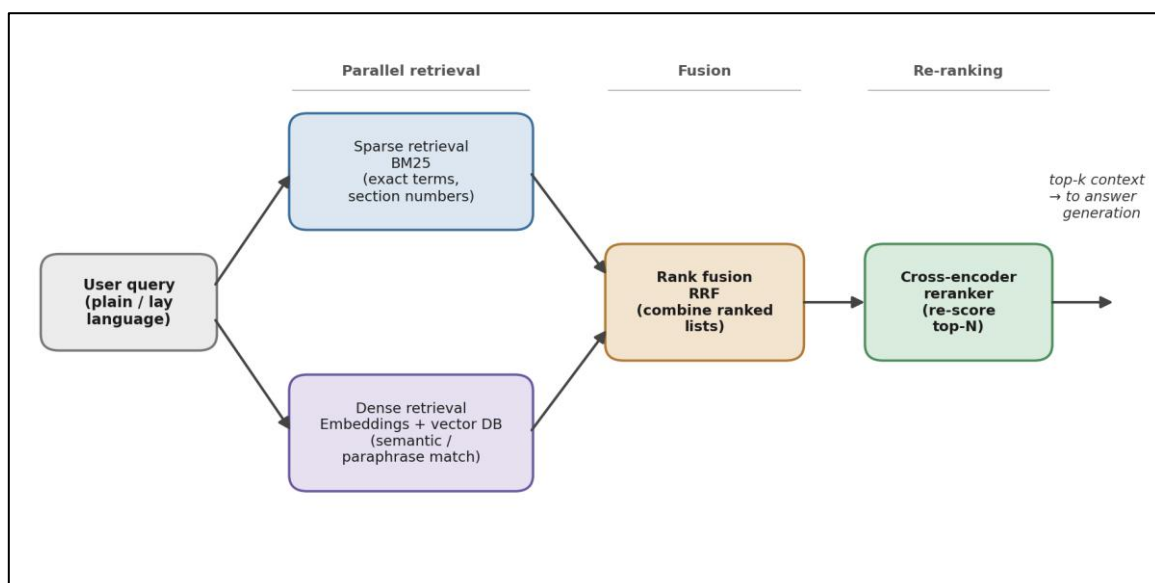


Figure 2. 3 Hybrid Retrieval with RRF and Reranker

Overall, the literature provides a clear retrieval architecture. With sparse and dense retrieval failed on complementary situations, hybrid retrieval approach combines through Reciprocal Rank Fusion and refined by reranker are consistently outperforming single retrieval methods in order to reduce hallucinations, which serves as the reason for the idea behind this project. The overall retrieval pipeline design is shown in Figure 2.3. At the same time, entire evidence is derived from either open domain or non-Malaysia Legislation environment, and the efficiency of reranker used in open domain may not automatically apply in legal context. The hybrid retrieval method that is able to capture query with plain language and technical term has not been tested or published on Malaysian statutory background, which is precisely what this project aims to investigate. Even with accurate section being retrieved, it does not guarantee that answer generated by LLMs will be faithful since the generation phrase may still include extra justification. With that, Section 2.6 will examine the verification mechanism strategy to ensure the generated output can be traced back to the actual section.

2.6 Citation Verification in RAG

Section 2.5 has concluded that hybrid retrieval method with reranker increase the accuracy of retrieving the correct statutory section based on query, but it does not guarantee that answer generated by LLM will be trustworthy. The difference is significant because the two different point of failure identified in Section 2.3 are separate issue. This suggests that even the right context is being retrieve, the generation stage could still generate output that is not supported by the context retrieved. This section therefore shifts the concern from what is retrieved to whether the generated answer is faithful by reviewing how the existing studies define and verify answer grounding, addressing the one of the two points of failure, which is faithfulness failure. Major legal research tools that use retrieval like LexisNexis and Thomson Reuters has been audited and the results reveal that the system often invent legal interpretation that cannot be traced back to legal documents (Magesh et al., 2025). This suggests that retrieval alone does not make the system trustworthy, especially for laypeople who is unfamiliar with the law themselves. Hence, a verification layer is treated as a necessary mechanism rather than optional for a reliable legal QA system.

2.6.1 Citation Recall and Precision

Liu et al. (2023) proposed that a trustworthy generative system should be able to verify its outputs against the actual sources, meaning that the output should be cite from original sources so that each of the statement can be supported by a citation. Furthermore, the citation is expected to be cited correctly so that each citation supports its associated statement. This study conducted evaluation test manually to check the quality of the outputs, turned out that many statement do not backed by any citation, many citations do not support the statement, and citation given is incorrect. These properties can be evaluated through citation recall and citation precision. T. Gao et al. (2023) proposed an Automatic LLMs' Citation Evaluation (ALCE) benchmark that evaluate the citation recall and precision automatically and revealed that even with the best models still have room for improvement on the citation quality. Together, these two metrics should be adopted by citation verification component of Malaysian statutory law RAG system since the citation recall and precision directly associate with whether a generated answer can be verified by the actual sections of the Act.

2.6.2 Citation Faithfulness

While citation recall and precision confirm that the citation exists and pointed toward the relevant context, it does not confirm that the generated answer's reasoning is based on that section. Research done by Wallat et al. (2024) shows that the generated answer can be cited by genuinely relevant section but the reasoning in the answer is not derived from the quote, and they have suggested that current generated answers are lacking citation faithfulness with up to 57 percent of the cases they explored. This insight is vital to this project because it means a verification layer alone cannot prove that citation section has existed or related, it must further confirm that the claim by LLMs is supporting by the cited text. Thus, it is highly important to evaluate the faithfulness for a reliable RAG system. Retrieval-Augmented Generation Assessment (RAGAS) framework introduced by Es et al. (2024) evaluates the faithfulness with no predefined answers written by humans. Instead, it decomposes answers into separate claim, and check whether each of the claim can be supported by the retrieved context. Since RAGAS evaluation only require the query, retrieved context and generated

answer by LLMs, the system is able to automatically determine whether the answer is supported by retrieved context rather than relying on human judgement. However, RAGAS recall and precision still require ground truth verification, meaning a test set of Malaysia statutory law questions still required even when using RAGAS.

2.6.3 Verification Architectures

Beyond evaluation, existing research propose different strategies to ensure the answer generated by LLMs is faithful through different degree of access to model's internal components. Asai et al. (2023) introduced SELF-RAG which the model is trained to critique on its own generation with retrieved passages using reflection tokens that embed the verification feature inside the generation layer itself. On the other hand, Qi et al. (2024) presented Model Internals-based RAG Explanations (MIRAGE) that assigns each generated statement to the retrieved context by analysing the model internal processing to identify which retrieved context is contributed to the generated statement. Although it is more accurate than self-citation, the researchers only used open-source LLMs like Zephyr and LLaMA since MIRAGE required full visibility into model's internal workings (Qi et al., 2024). This indicates that commercial Application Programming Interfaces (APIs) like Claude or ChatGPT could not be used with MIRAGE frameworks, as it requires access to model internal architecture. Thus, it is not suitable for MIRAGE to adapt in this project. As a results, the proposed system in this project must rely on verification techniques that compare the generated answers with retrieved section without the needs to interpret the operation in the model.

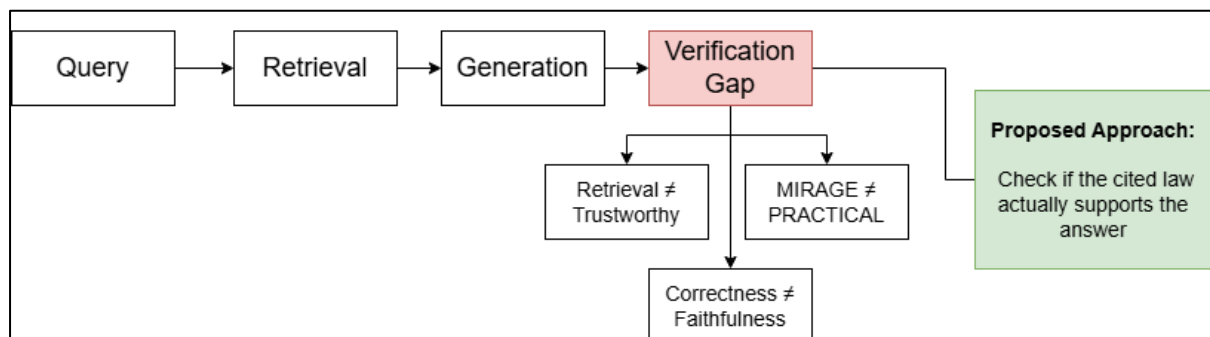


Figure 2. 4 Verification Gap for RAG system

Ultimately, this section explores three points that shape the proposed architecture in this project which is summarize as verification loop for RAG system as shown in Figure 2.4. First, generation without any citation verification may still undergo output being unsupported or hallucinated citation. This makes the verification is necessary. Second, correctness of an answer does not make it faithful, so a reliable architecture must check whether the answer is supported by the cited section, rather than only checking whether the citation is present. Lastly, MIRAGE requires access to internal component of model which therefore inapplicable to close source APIs generation even though the studies indicated that it is more faithful to the model actual reasoning. Thus, a more practical alternative is used to verify answer from retrieved section directly. What remain absent from all the research is these mechanisms do not evaluate on Malaysian statutory law where all the benchmark and method was developed on open domain data or non-Malaysian legal system. This regional gap will be further analysed in Section 2.7 which review existing work on RAG for Malaysia or Asian regional legal system.

2.7 RAG for Non-English and Southeast Asian Contexts

Section 2.2 to 2.6 has discussed how the hybrid retrieval approach and verification mechanism perform in legal question answering system. However, most of the research were conducted or developed using open domain data, United States case law and other Europe statutory law. This section redirects the discussion to non-English based and Southeast Asia legal system to explore whether the techniques discussed earlier can be applied in these regions. This concern arises as legal language and structure of statutory law is different among countries. Therefore, approach that function effectively in existing statutory law RAG system may not automatically transfer directly to Malaysian statutory law.

2.7.1 Non-English Legal Context

Beyond Western English jurisdictions, several studies that were conducted using language other than English. Louis and Spanakis (2022) presented the Belgian Statutory Article Retrieval Dataset (BSARD) with over 1,100 data of French legal

question derived from over 22,000 of Belgian law articles. The findings of this research indicate that fine-tuned dense retrieval models outperform other systems, with only achieving 74.8 percent Recall@100, showing that there is still have room for improvement. This proves that even when using languages that have large legal datasets, retrieving relevant section from statutory documents is still imperfect, showing that it is an unresolved problem rather than solved one. Alternatively, Nguyen et al. (2024) proposed an attentive neural network-based approach to retrieve statute law and were evaluated through Japanese, Vietnamese and English language dataset which surpassed non-neural strategy by achieving the highest recall. It demonstrates that statutory law retrieval can be performed in Asia with low prior legal resources. However, this pipeline does not include any generation or citation component, implying that retrieval alone could not solve the problem of user who require a direct and faithful answer rather than interpreting themselves with a ranked list of section retrieved.

2.7.2 Southeast Asian Legal Context

Within Southeast Asia environment, Faisal et al. (2024) built a question answering legal system with BM25 and BERT as retriever and a reranker on 15 government regulations of Indonesia across several component granularity levels like chapters, articles, sections and letters. It is highly relevant to this project as it utilizes the “retrieve then rerank” approach to improve the accuracy from 44.86% to 63.24% while using the section chunking strategy that this project will adapt. While it is evaluated on broad sets of regulations, it only returns the correct components rather than generating natural language answer and a citation verification mechanism, meaning the faithfulness of the output that emphasized in Section 2.6 still remain unknown. Yulianti et al. (2024) on the other hand, introduced IndoLER dataset that consist of approximately 1,000 of court decisions and were tested on several models to extract important information from documents, also known as legal entity recognition (LER) tasks, with XLM-RoBERTa models identified as the best performing model with F1 score of 0.9295. Although it is an entity recognition rather than RAG, it provides further evidence that research of Natural Language Processing tasks in legal domain is active in Southeast Asia setting but mainly focus on Indonesia instead of Malaysia.

Niklaus et al. (2023) spot the gap of many benchmarks were only made available in English, so they presented LEXTREME, a multi benchmark that consist of 11 datasets from legal domain across 24 languages. While it shows that multi language legal NLP is accepted as key area in research, the paper uses language mainly from Europe legal system and did not contain any Asian or Malaysian statutory jurisdiction. This further highlight the research gap which Malaysian or Southeast Asia were not represented even though the benchmarks were designed specifically to extend the language for NLP.

2.7.3 The Malaysian Gap

Looking at Malaysia specifically, there is currently lack of research on RAG for Malaysian statutory law. The only study done by Khong and Ho (2022) is closely related to the AI used in sentencing application in Sabah and Sarawak courts. It uses machine learning algorithms for criminal sentencing and court processing rather than a question answering RAG system. Lack of research in retrieval-augmented generation to Malaysian legislation showing a gap where the combination of hybrid retrieval and citation verification has not been discovered in existing research.

In summary, this section has showed that the legal NLP is actively developed in European and Indonesia but none of them were conducted in Malaysian statutory law. Existing work from Southeast Asia mostly stops at the process of retrieval, without the generation phase and some of the studies were focus on other tasks that still falls under the NLP umbrella like LER, while AI research in Malaysia is limited to assisting judges and judicial officers rather than a system that is developed for citizens. This confirms that there is a critical gap on the Malaysian statutory RAG legal system that will be discussed further in Section 2.9.

2.8 Comparison of Existing Work

Earlier section discussed about the question answering in legal domain and introduced the RAG (Section 2.2 and 2.3), existing RAG used in legal domain (Section 2.4), the hybrid retrieval and reranking approach (Section 2.5), citation verification in RAG

(Section 2.6) and finally the usage of RAG in legal domain for non-English and Southeast Asia context (Section 2.7). This section summarises and compares the existing systems so that their strength and limitation can be used to justify the choice of architecture for this project. Table 2.1 summarizes the comparison with its law type, retrieval method, the presence of citation verification, key finding and its gap relative to LexRAG-MY.

Author & System	Law type/ Dataset	Retrieval Method	Verification mechanism	Key Finding	Gap relative to LexRAG-MY
(Magesh et al., 2025) LexisNexis	US case law	Not disclosed retrieval method	None, audited result revealed that system may produce false information	Legal RAG system still hallucinates around 17% to 33% of the time	Conducted on US case law, Motivate verification strategy approach
(Hou et al., 2025) CLERC	US case law	Dense + BM25	None, hallucination rate is evaluated but not prevented	BM25 Recall@1000 only 48.3%	Conducted on US case law, no verification step to reduce faithfulness
(Louis et al., 2024) LLeQA	Belgium and France statutory in French	Comparing Dense with BM25	Answers include citation but is not used for tracing back to the actual context	Both retrieval method performs well in automatic evaluation test, but error was found when human review	Uses French language, separate pipeline was used rather than hybrid approach
(Wiratunga et al., 2024) CBR-RAG	Australian Open Legal QA (ALQA) dataset	Dense + BM25	None, retrieval level only	Hybrid retrieval performs better than single retrieval	Not in Malaysia legal context, do not have a generation layer to access the direct and

Author & System	Law type/ Dataset	Retrieval Method	Verification mechanism	Key Finding	Gap relative to LexRAG-MY
					faithful answer
Li et al. (2025) LexRAG	China statutory law	Dense + BM25	Yes, evaluated faithfulness	Proposed idea of evaluating faithfulness and completeness in statutory law	Chinese language document was used
(Faisal et al., 2024)	Indonesian jurisdiction	BM25 + BERT reranker	None, only return sections without generating natural language response	Retrieve then rerank approach improves the accuracy from 44.86% to 63.24%	Only uses sparse retrieval with no generation layer and verification layer
(Louis & Spanakis, 2022) BSARD	Belgium Statutory	Comparing fine-tuned dense retrieval with sparse architecture	None, only retrieval tasks	Dense outperforms sparse retrieval, but the best result is only achieving 74.8 percent Recall@100	Tested on separate pipeline, with no generation and verification layer
(Nguyen et al., 2024) Attentive CNN	Statute law document in Japanese, Vietnamese and English	Attentive neural network-based approach	None, evaluated recall and F2 for retrieval only	Neural retrieval works in low resource legal language	No generation steps to verify the faithfulness of output
(Khong & Ho, 2022) AiCOS	Offenses for drug possession and rape in Malaysia	None, it is ML rather than a RAG	None, not a RAG or QA system	The only research found on AI used in Malaysia legal domain	Used for court procedure, not a QA system for Malaysian statutory law

Table 2. 1 Comparison of Existing Work

The coverage that listed in Table 2.1 shows a consistent pattern of weakness which maps each of their key finding against the component of LexRAG-MY. Each reviewed studies contributed to one or two components, but none combine all of them, and the Malaysian statutory settings in RAG system remain insufficient. This highlights that there is lack of research in the hybrid retrieval with a verification mechanism for a QA RAG system for Malaysian statutory law. Thus, there is a research gap which will be analysed further in the next Section.

2.9 Research Gap

Section 2.8 has discussed and summarized the existing system through Table 2.1 with each of them having consistent weakness among the reviewed studies. This section combines all those observations into research gaps that this project addresses. Based on the comparison, three gaps can be identified throughout the literature which is the legal system gap, retrieval method gap and verification gap.

First is the legal system gap. As shown across the literature, most of the research were being conducted on United States case law like CLERC and LexisNexis, or non-Malaysian statutory law such as LLeQA, BSARD and other Asian country like Indonesia and China law. Section 2.7 revealed that not a single of research performed retrieved augmented generation on Malaysian statutory law, with only closely related study by Khong and Ho (2022), but it is a sentencing tool for Sabah and Sarawak court rather than a statutory QA system built for citizens. This matters as the structure and legal terminology used between countries are different as discussed in Section 1.1, which the methodology and result cannot be automatically transfer directly between them.

The second is the retrieval gap. Section 2.5 concluded that hybrid retrieval that combine sparse with dense retrieval method, merge them using RRF along with a reranker outperform single retrieval method. This evidence come from Wiratunga et al. (2024) demonstrated that hybrid retrieval performs better and Faisal et al. (2024) reported that “retrieve then rerank” strategy improved approximately 20 percent of accuracy. However, Table 2.1 shows that most of the statutory law in Southeast Asia

is retrieved on single retrieval method, leaving the full hybrid method with a reranking strategy remain unknown in Southeast Asia statutory context.

The last and most critical gap is the verification gap. The verification mechanism remains one of the core features in this project, but the column for verification mechanism in Table 2.1 is almost empty, with only 1 out of 9 articles evaluated the faithfulness of the generated answer, that is LexRAG by Li et al. (2025). However, it is being designed in Chinese language environment, therefore limiting its use in non-Chinese languages. This confirms that research on citation verification approach that validate the faithfulness of the generated answer by LLMs remain insufficient in existing legal RAG system.

Collectively, each study closed only one or two of these gaps, but none combined hybrid retrieval with citation verification in Malaysian statutory law settings. The present of these gaps motivates and shapes the objectives of this project that develop a retrieval strategy that is capable of locating the accurate parts of documents based on plain language legal questions (O2), develop a citation verification system that verifies the accuracy of the output from LLMs by checking the generated answer with the retrieved source section rather than produce extra interpretation or assumptions (O3) and evaluate both components using test set of Malaysia statutory law (O4) to address the legal system gap. Section 2.10 conclude this chapter by summarizing the key finding of literature review.

This chapter analysed the existing research that support the development of this project with its research gap that aim to address. The review starts by analysing the use of AI and the development of question answering system in legal domain in Section 2.2. It concludes that QA system for legal domain is more complicated than general purpose QA as it not only requires to generate output accurately but also ensure that statement can be traced back to the sources. Evaluation method like LegalBench and LLeQA existed but neither of one covered Malaysian statutory law. Section 2.3 then introduced the concept of retrieved augmented generation by explaining the process of retrieving passage from documents before generating answer which combine both non-parametric and parametric memory to retrieve content from external documents and generate output based on the context, while

Section 2.4 narrowed down the discussion into legal domain and confirm that RAG can be used to lowered but not fully eliminate hallucinations.

The literature review then moved towards the technical and choice of pipeline that can be used for RAG. Section 2.5 discussed the hybrid retrieval that combine sparse and dense retrieval method through Reciprocal Rank Fusion and rerank them to ensure only the most accurate context are being pass to LLMs and concludes that hybrid retrieval method outperforms single retrieval methods. Section 2.6 analyse the citation verification approach with faithfulness in RAG and highlights that a fluently generated answer might not be faithful unless it can be trace and verify back from the retrieved sources rather than relying on model's own interpretation. Section 2.7 then compared the RAG used in non-English and Southeast Asian country's legal system and found out that most of them were develop using only single retrieval method, with lack of research conducted on RAG used in Malaysia statutory law.

Section 2.8 compiled all the reviewed work into a table that shows a consistent weakness across 3 gaps that were identified in Section 2.9, which is the legal system gap, retrieval gap and verification gap. These gaps confirm that there is lack of existing research on hybrid retrieval with a citation verification in Malaysian statutory law which shapes the objectives of this project and influence the system design that will be introduces in the next section.

Chapter 3: Methodology

3.1 Introduction

Chapter 2 have reviewed the existing research and concluded with 3 research gaps, which is the legal system gap where most of the study conducted on United States case law with lack of research focused on retrieved augmented generation for Malaysian statutory law, the retrieval method gap where almost all of the analysis in Southeast Asia statutory law were performed using only single retrieval method, and the verification gap where only limited studies examined on validate the faithfulness from the generated answer which is fluent but may not be faithful. The literature also named the strategy that will be used, which is the hybrid retrieval method that merge the retrieved section using Reciprocal Rank Fusion (RRF) with a cross-encoder reranker, and a citation verification layer to confirm that the generated statement can be traced back to the retrieved sources. This chapter then convert the choices of the design into methodology.

The purpose of this chapter is to describe the design, implementation and evaluation of the proposed system, so that the second, third and fourth objectives can be met. Objective O2 will be address in Section 3.4 that explain the retrieval pipeline of getting relevant passage of Malaysian legal documents based on query. Objective O3 will be achieves in Section 3.5 on the constrained of the answer generation and in Section 3.6 on the verification mechanism. Lastly, objective O4 will be discusses the evaluation method in Section 3.7 that compare the full proposed pipeline against simpler retrieval method using a self-built test set of Malaysian statutory questions.

This chapter starts from introduction in current section, then Section 3.2 introduce the overall architecture of the system design. Section 3.3 explain the preparation of dataset, including the selection of Malaysian legal documents with the chunking strategy. Section 3.4 focuses on the retrieval pipeline and Section 3.5 explain how the answers will be generated. Section 3.6 reveal the verification mechanism with its error handling features, follow up by Section 3.7 focus on the evaluation method which include the generation of test set and metrics for retrieval and generation components. Section 3.8 list down all the tools will be using throughout the whole project and Section 3.9 discuss the potential ethical issues.

3.2 Overall System Architecture

This project uses the Design Science Research (DSR) approach where it is constructed to solve specific problem and evaluated through measurable metrics to ensure the effectiveness of the system that actually solve an issue rather than simply claiming it as a useful tool. To evaluate it, only the retrieval method will be changed while the remaining component like answer generation and evaluation dataset will remain the same. The purpose of this decision is to answer the research question Q4 that ask whether the proposed retrieval and verification system will produce more reliable output as compared to simpler RAG pipeline. This ensures that the system performance can be compared in a controlled environment rather than doubting on which component leads to the results. The pipeline that will be develop and evaluate in this project named LexRAG-MY, with the phrase “Lex” means “Law” in Latin, and MY used as abbreviation for Malaysia. Putting it together represents the use of RAG in Malaysian legal domain.

The pipeline is divided into 2 architectures. First is the offline stage (Stage 0) that only require to be run once. Two of the Malaysian statutory Acts, which is the Employment Act 1955 and the Consumer Protection Act 1999 will be download from Attorney General's Chambers of Malaysia as PDF and convert into clean text, split it into chunks based on the Sections, enhanced with metadata and index twice for both keyword retrieval and dense semantic retrieval in the offline stage. Another architecture is the online stage that will be trigger each time user asking a question. Plain English query will be pass through the hybrid retrieval with RRF fused the results and reranker enhanced the results, select the top-k most relevant section and LLMs will generate answer strictly based on the retrieved context. Then, it will be verified it through existence and support check before displaying the answer to user. This system support error handling when not all the claims were supported from the retrieved section. It will retry with single stricter regeneration and refuse to answer if it fails again rather than providing assumption answer. The full architecture of this design is illustrated in Figure 3.1.

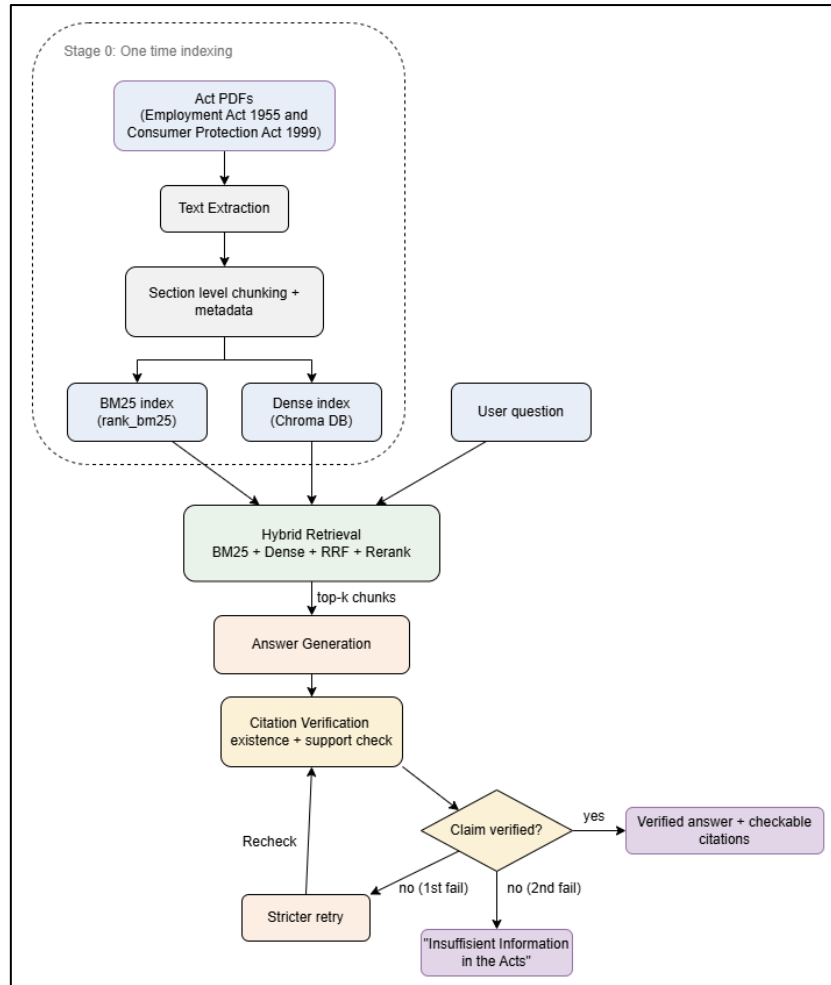


Figure 3. 1 Full Architecture of LexRAG-MY

The architecture is modular that can be replaced or removed for every component, without rewriting the entire application which aligns with the Modular RAG that discussed in Section 2.3. Each component has its own input and output, so it can be replaced without affecting the flow of the pipeline. Even though the benefit of modular allows the system to change within specific components without affecting the entire system, the main reason for the modularity in this project is to answer the research question RQ4 where the retrieval type must be able to swap while keeping other components remain identical so that evaluation in Section 3.7 can be done. Thereby, evaluation of different retrieval approaches can be observed with controlled environments under this condition.

Several computing factors were influenced by the architecture design. In terms of its performance, both retrieval methods will be only indexed once and reused, as shown in Stage 0 of Figure 3.1 so that the system can search from existing indices instead of reprocessing the whole documents again when a user asks a question. This enables

faster response time and less computational effort for each question asked. From the usability perspective, LexRAG-MY is design for Malaysian layperson that enable user to ask questions in natural English language so that they do not need to understand the complex terminology, which addressed one of the problems in Section 1.2. For example, a user can simply ask “Can I get any refund for receiving defect item?”, rather than asking in legal language like “What constitutes a false or misleading representation under the Consumer Protection Act 1999?” or “Please explain Section 32 of the Consumer Protection Act 1999”. For reliability and safety, this system ensure that all generated answers can be tracked back to retrieved section and no answer will be generated if the system failed to find the evidence in the Acts rather than providing assumptions. Regarding its reproducibility, the exact version of legal Acts and the configuration of system will be documented for future researchers to rebuild the same system and set as baseline for Malaysian statutory legal RAG system. LexRAG-MY can be operated in two modes using the same pipeline. User mode that is intended for end user which allow user to ask question and receive a natural language response in real time. Batch mode allows set of questions passed through the pipeline at the same time for system evaluation and record metrics like recall and precision based on the outputs. The comparison different operating mode is shown in Figure 3.2.

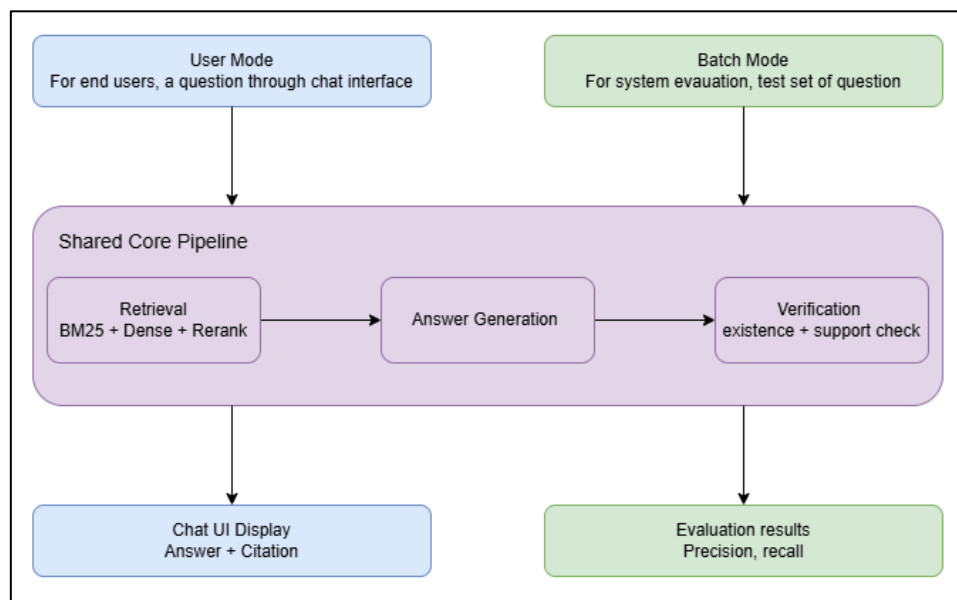


Figure 3. 2 User Mode and Batch Mode with a Shared Core Pipeline

3.3 Dataset Preparation

As discussed in Section 2.3, the efficiency of RAG system is highly dependent on retrieval quality, so the preparation of dataset is crucial and should be treated carefully rather than handling it as a basic preprocessing task. This section describes the two Malaysian Acts along with the extraction process and the section-level chunking strategy that will be used in the system in retrieval process.

3.3.1 Source and Selection of Acts

The dataset consists of two different Malaysian statutory law that were established by Malaysian Parliament to address Malaysia's social issues, which is the Employment Act 1955 (Act 265) and Consumer Protection Act 1999 (Act 599). Both documents were publicly accessible source that can be view and download in Attorney General's Chambers of Malaysia at <https://www.agc.gov.my/>. Using the official sources rather than third party platforms ensures that statutory documents contain original, accuracy and up to date information which address the main purpose of this system, that is providing user with a reliable answer.

Only two Acts are select instead of the whole corpus of Malaysian statutory legal document. This allows the cleaning process and the construction of test set that require a lot of manual effort can be completed within the proposed timeline. Additionally, the two specific Acts were chosen, particularly employment and consumer protection that may affect Malaysia ordinary people that cover different area of law, allowing the system to answer wide range of questions without require large amounts of documents. Furthermore, both documents are statutory law instead of case law as the structure of the document makes them suitable for retrieval methods that can be organized by parts and sections. The version of the document and download date from the official sources will be documented because the document may be amended and differ from each version. This ensure the reproducibility of the system where test set of the question remain consistent with the indexed text from the exact version of the documents.

3.3.2 PDF Parsing and Chunking

The two downloaded documents are published as PDF files, which the texts need to be extracted before they can be split into sections. Text extraction is performed using PyMuPDF, a python library that allow documents to be read page by page for extracting and manipulating PDF documents. Since some of the output extracted may not be relevant to the context, example the header and page number, a further cleaning process is required so that only the actual content from the Acts remained. The cleaning process will be performed based on the consistent layout of documents and will be manually checked to ensure the accuracy of content.

While text extraction is a necessary step to ensure content of documents can be further processed, the decision on how the cleaned text can be split into chunks remain the most important decision in this stage because the chunk of document will be use as context for input and output of retrieval, generation and verification stage. Instead of splitting text into constant chunks size that a general-purpose RAG system would use, the text extracted from the sources will be split according to structure of the Acts, producing one section per chunk with its section number as unique identifier like Section 60D(1)(a). The section chunking strategy is adapted due to the structure of Malaysian statutory law where each section contains a complete legal information. In contrast, fixed length window chunking will split similar information into different chunks, which lead to poor citation quality because retrieved chunks are not matched with the complete section. Figure 3.3 illustrates how a single section can be split into chunks with each section carry their own unique identifier.

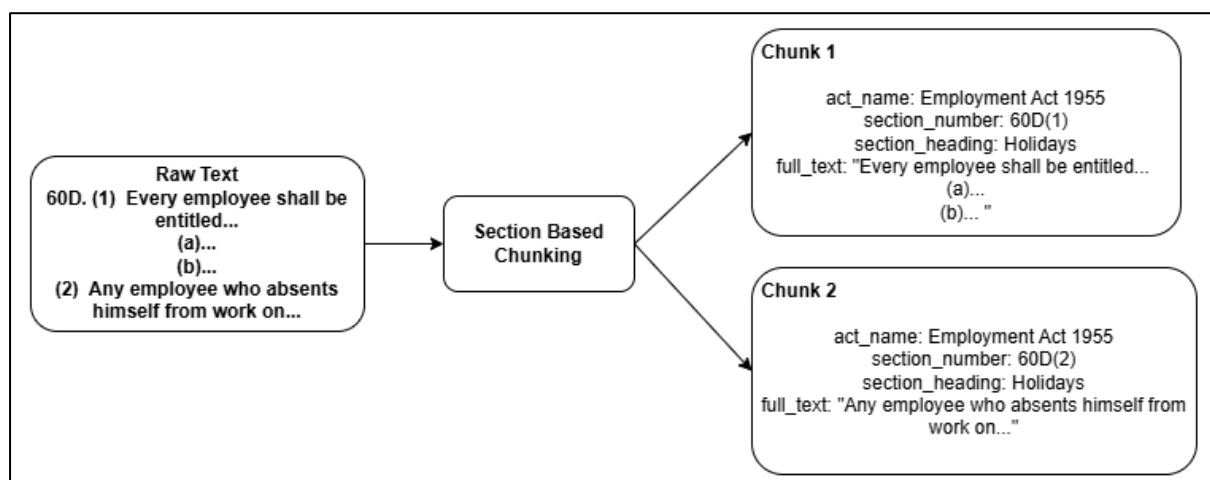


Figure 3. 3 Example of Section Chunking Method using the Employment Act 1955

Each chunk will be stored with the metadata that record the identifier and content from the chunk. The table and column name of metadata is summarized in Table 3.1. The purpose of this metadata is to ensure that the generated answer can be traced back with the content from the identifier, and also to verify whether the cited section from LLMs were genuinely among the retrieved sections in verification stage.

Field	Description	Example
act_name	The document name	Employment Act 1955
section_number	The identifier of the chunk	60D(1)(a)
section_heading	The title of the section where chunk belong	Holidays
full_text	Full statutory text from the section_number	Every employee shall be entitled ...

Table 3. 1 Metadata for Statutory Chunks

3.4 Retrieval Pipeline

The retrieval pipeline is part of system that objective O2 address which is to design a retrieval approach that able to accurately retrieve relevant section based on question that is phrase in non-technical English language. As discussed in Section 2.5, the two retrieval method serves as complementary strength rather than direct substitute where sparse retrieval is effective in capturing the exact legal terms and identifier but fail to match when user ask in plain language. Dense semantic retrieval solves the issue by understanding the meaning of the sentence but may overlook the technical term or unique phrase. LexRAG-MY combines both methods using RRF and enhance it through a reranker. Full pipeline along with three other retrieval option used to solve the research questions Q4 for evaluation is shown in Figure 3.4.

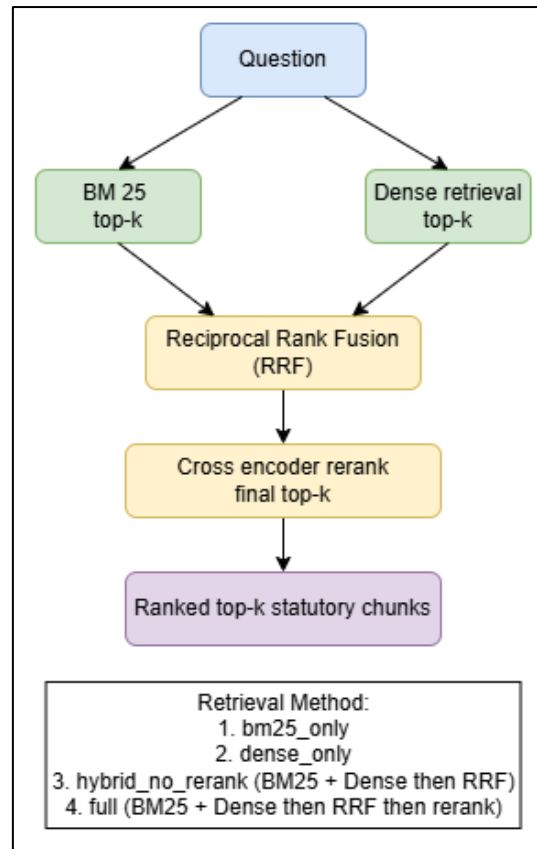


Figure 3. 4 Hybrid Retrieval Pipeline

The pipeline consists of 4 different types of retrieval configuration so that each stage can be evaluate and measure. The bm25_only retrieval configuration uses only sparse keyword search while dense_only uses only semantic search. The hybrid_no_rerank combine both retrieval method and merge them into single retrieved list of chunks through Reciprocal Rank Fusion. In the full configuration, the proposed system, it enhances the hybrid_no_rerank method by adding a reranker before the final top-k chunks are selected. Each configuration returns a list of top-k statutory chunks that will be feed as context for LLMs in generation stage.

3.4.1 Sparse Retrieval

Sparse retrieval method is implemented with rank_bm25 python library. The calculation of relevance score is based on 3 different factors. Term Frequency (TF) calculate the exact term found within the chunks, Inverse Document Frequency (IDF) weighed the uncommon phrase with higher importance so it reduces the influence of common terms, and Length Normalization normalize the chunks length to ensure that

longer chunks do not gain advantage just because it contains more wording. The characteristics of sparse retrieval ensure the effectiveness of capturing the accurate section that rely on exact wordings like “Section 32(1)” which semantic retrieval might treat it as numbers with meaning embed with it or may even treat it as interchangeable similar looking numbers. For each chunk, the full text will be tokenised once and added into the BM 25 index during the Stage 0, as shown in Figure 3.1, whereas question is tokenised during the online stage when user asks a question and compare the score against all chunks with the top relevance chunks are returned. The properties of BM 25 despite its limitations of semantic search are exactly the complementary strength that discussed in Section 2.5 where dense retrieval may miss the technical legal terms or rare phrases.

3.4.2 Dense Retrieval

Dense retrieval is designed and used in this project to understand the meaning of the question even when the question does not contain any exact wording that match the statutory documents. Each chunk will be converted into numerical vector through embeddings where the chunks are located near to each other when they have similar meaning even though no exact word were matched. These vectors will only run once during the offline phase, and will be store in ChromaDB, an open-source vector database designed to store the embeddings. Every time when user ask a question, that question will also be converting into embeddings using the same model. Then, it will return top-k lists of chunks which the numerical vector is closest to the embeddings of user question through the cosine similarity. This allows the system to understand the semantic meaning of question from user and retrieve the accurate chunks of section even when the user wording is different from the legal document. Due to the cost constraint of external APIs will be charge when processing each chunk of documents, and the time constraints where each indexing process leads to network latency when transmit data through an external embedding service, the generated embedding vectors will be store in ChromaDB locally for the retrieval component of LexRAG-MY.

3.4.3 Hybrid Fusion (Reciprocal Rank Fusion)

Once sparse and dense retrieval produce its own top-k ranked list, it must be merge and combine into a single ranking. This can be done using Reciprocal Rank Fusion (RRF) that is being introduced in Section 2.5. For each chunk in the ranked list, RRF calculates the rank score using the formula below so that the chunk that constantly stay on top of both lists will receive a higher combined score.

$$RRF(d) = \sum_{i=1}^N \frac{1}{k + rank_i(d)}$$

Where:

- N = number of retrieval system
- $rank_i(d)$ = chunk's rank in that list
- k = constant (commonly 60)

RRF is chosen rather than other fusion techniques because it works well even with very different type of retriever. In this project, both BM25 and dense retrieval produce different types of relevance scores which cannot simply merge them. RRF solve that by ignoring the score and only considering the ranking position, rather than performing score calibration technique to normalize the relevance score into the same scale. Furthermore, RRF simply uses a mathematical rule to rank the chunks and do not require past data and hence no training is required. Thus, RRF is the correct approach for the project since currently there is lack of preexisting labelled question dataset on the Malaysian statutory context.

3.4.4 Reranking (Cross-Encoder)

While fusion combined retrieval method into a single ranked list based on mathematical rule, the ordering can be further sharpened in the reranking stage. The top-k of chunks from fusion will be pass to cross-encoder reranker which reads the question and chunk as a pair and produce a relevance score for each of the pair. Reranker compare the relevancy of the chunks direct with the question, so it can produce a more accurate ranked list as compared to RRF. However, reranker increase the computational cost as it needs to compare every single pair of question and chunk to produce their own relevance score. Hence, the “retrieve then rerank” steps that

discussed in Section 2.5 is adapted in this project where only small set of candidates will pass through the reranking stage rather than all available chunks, which is exactly why the reranker component is placed after the hybrid retrieval method rather than used it as a primary retriever method before handed to the LLMs as contexts. Additionally, Section 2.5 has also questioned the accuracy of reranker because it was trained on the open domain data which the effectiveness does not automatically apply in Malaysia legal context. This project does not automatically admit the effectiveness of using reranker in Malaysian statutory RAG system, but rather another possible architecture that can be used to compare or produce a more effective output between the available retrieval architecture that address the research objective O4.

3.5 Answer Generation

After the retrieval component selected the top-k statutory chunks, it will be feed into LLMs to generate a plain and non-technical language. The answer generated should allow user to verify against the actual retrieved context which address objective O3 that prevent hallucination answers. The generator, also known as large language models (LLMs) will be accessed through external APIs that allow the system to send prompt to models like OpenAI, Claude, DeepSeek or more and receive responses from them. However, their output will be constrained to ensure the model only explain strictly based on retrieved section rather than proving extra assumption based on their training knowledge.

3.5.1 Prompt Design and Context Constraint

The retrieved chunks from reranker will be assembled into single prompt under a set of instruction to ensure answer generated were strictly based on the context. The RACE Prompting Framework is used in this project when designing a prompt, which consists of Role, Action, Context and Expectation. Role specifies the model to act as a legal information assistant before provides any context. Action tells the AI exactly what tasks to perform which set the scopes and constraints for the model, while context provide the retrieved section to complete the action, shifting the question from “what do AI expect to do” to “what material is provided to AI”. Lastly, expectation tells

the AI on what exactly the output is should look like, which in this system make sure that each claim of the generated answer must be supported and cited by the exact section to ensure the factual accuracy. This output support manual checking from the Acts. Additionally, the output from LLMs will not provide explanation or reasoning over user personal scenarios. Instead, it only explains the relevant section that possibly combining multiple sections into single explanations, retrieved chunks, and a system prompt to ensure the system stay within the safe boundaries which prevent assumption is made based on user's personal scenario. However, if the provided section does not answer the question, the model should reply as "insufficient information based on provided Acts" and do not provide any legal answers. This ensures that model refuse rather than fabricate answers, preventing hallucinations and improves the faithfulness of the response. By only supplying retrieved chunks into prompt and restrict the use of model's own training knowledge, the system aims to only generate answer based on what is in the context where each statement must be traceable back to specific chunks. This makes the tasks of model solely to understand user questions, combining multiple chunks from technical legal term into single natural language translation based on the question, and following the constraints that were set.

3.5.2 Citation Generation

The model is expected to return generated answer in a structured form rather than free text which consist of an answer based on user questions, and also a list of individual claims that include the claim text based on the answer and the cited section it retrieved from. By doing so, it enhances the verification process and making it simpler and more reliable since each claim is already linked to its source, the verifier can directly check each statement with the cited section instead of separating the free text from model into individual claims and search for each retrieved section based on individual claims. Furthermore, producing a standardized format of output also ensure that system can always extract the claims and its citation in a predictable way rather than producing a free text with another layer separating the answer and its claims. This allows the claims to be pass down automatically to the verification component without further analysing free text paragraph. However, LLM may not produce a structured form as it does not create a guarantee programming output. Hence, a

schema checker will be implemented to ensure the output follows the required structure. Otherwise, the system sends the output back to the LLM again with stricter instructions and triggered fallback handler that display “Unable to generate a verified response.” if the generation still fails to provide a structure format.

3.6 Citation Verification Mechanism

This section, the citation verification mechanism, achieve the objective O3 that separate LexRAG-MY from Naïve RAG to a Modular RAG pipeline. The role of this component is to ensure that each claim generated by LLM can be genuinely traced back from the retrieved section, ensuring the faithfulness of the system so that user could receive answer that is reliable and allowing traceability from actual source rather than simply believing it because the answer sounds authentic. Section 2.6 has explained the necessary of this mechanism because the citation exist may looks relevant to the generated answer, but the reasoning of the answer might not be derived from the retrieved section. Furthermore, Section 2.6 also discussed technique that require to gain access on model’s internal architecture like MIRAGE is not suitable in this project since the generator adapting in LexRAG-MY uses a closed-source external API. Therefore, the verification component compares the generated claims with retrieved section text directly without the need to interpret and fine tuning the model’s internal operations. Full mechanism of citation verification along with its failure handling procedure is shown in Figure 3.5.

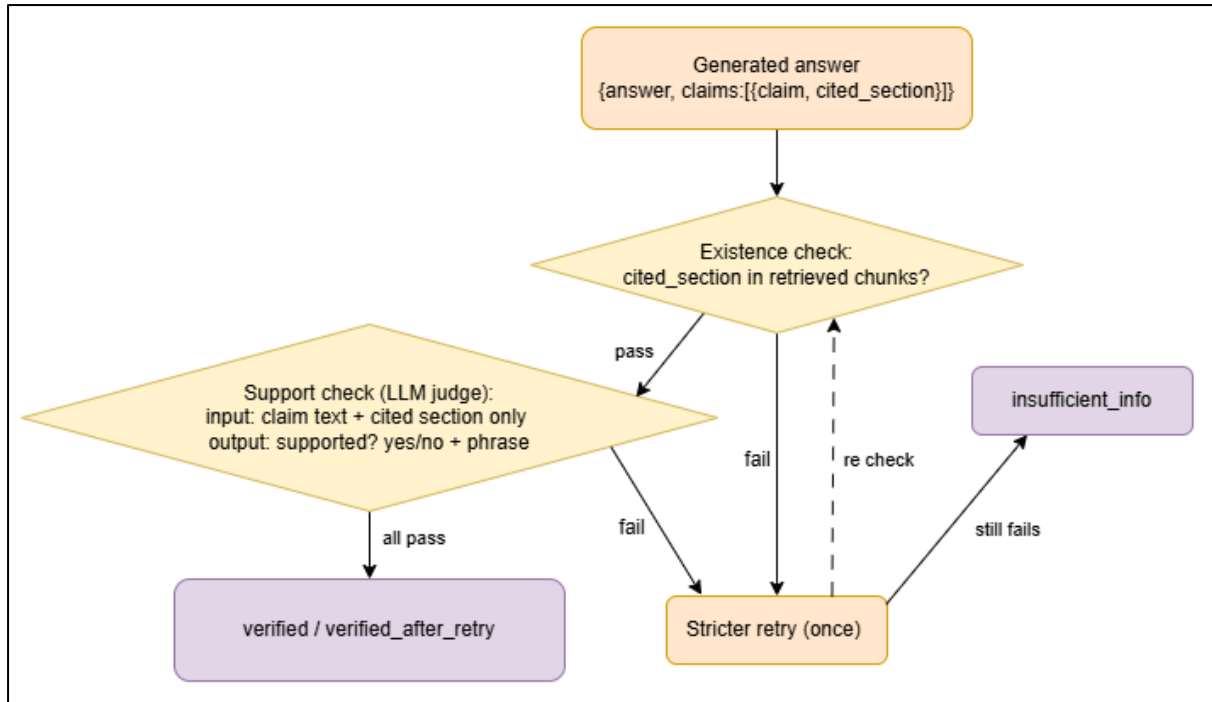


Figure 3. 5 Citation Verification and Retry Flow

Verification process strategy involves two checks. The existence check mechanism confirms that the generated citation is actually present in the set of chunks given by retriever that was feed into generator. If the cited section is not listed within the set of chunks, the claim fail immediately as LLMs might invent or forgot the identifier of the chunk. The second part of citation verification process is the support check where it uses the large language model as a judge. It only checks the relevancy of generated text with the cited text without providing the user’s question and rest of other statement that were cited by other sources. Support check asks whether the cited section support the generated text, with either output of “yes” along with the exact support phrase, or “no” along with statement stating that no supporting phrase exist. The constraints were set to prevent the judge from interfering by surrounding factors such as user question and prevent it from relying on LLM’s own trained knowledge. With that, the LLM used in support check is to make judgement and identified is the cited section actually supports the generated statement. The different between existence check and support check are compared in Table 3.2.

Aspect	Existence check	Support check
Main purpose	Verify the generated section actually exist in retrieved section	Verify the retrieved section actually supports the generated answer

Aspect	Existence check	Support check
Question answered	Did the model cite can be found in the retrieved section?	Does retrieve section actually prove the claims?
Type of check	Rule based	LLM as judge
Risk address	Prevent citation hallucination	Ensure faithfulness of the answer

Table 3. 2 Comparison of Existence Check and Support Check

Fault handling in this mechanism is designed to ensure that hallucinate or unverified answers is never displayed. If the generated answer does not pass through any of the check, the system will regenerate another output with a stricter prompt that explain only based on information stated in the retrieved section. Then, the second generated answer will still require undergoing both checks again where answer will return with status “verified after retrying” if it passes through both checks and otherwise return output that states “insufficient of information in the Acts”. The difference between initial prompt and stricter prompt are compare in Table 3.3. Thus, three types of status of output are “verified”, “verified_after_retry” and “insufficient_info” along with a citation as checkable reference are produced to ensure the verifiability of the system if it provide an answer. LexRAG-MY only allow a single retry of answer generation to balance between the answer coverage and safety which give the model another opportunity to produce a faithful answer but does not allow system to retry indefinitely to prevent hallucinate or produce an unsupported answer.

Aspect	Initial prompt	Stricter prompt (Regeneration)
Main instruction to LLMs	Explain using provided section	Explain only based on information stated in the retrieved sections
Goal	Create a helpful and natural answer	Create answer that is fully supported by sections
When to use	First answer generation	Only if first generation fail to pass through either check

Table 3. 3 Comparison of Initial and Stricter Generation Prompt

3.7 Evaluation Design

The objective of evaluating the proposed system is to address the objective O4 by measuring the effectiveness of this system on how well it can retrieve the correct sections and generates a faithful answer rather than assuming a RAG system can simply reduce the hallucination rates. Additionally, this section also compares between

full proposed RAG pipeline with simpler baselines. The evaluation is conducted using a self-built Malaysian statutory law test set by comparing the four retrieval design as defined in Section 3.4. The full evaluation flow is illustrated in Figure 3.6.

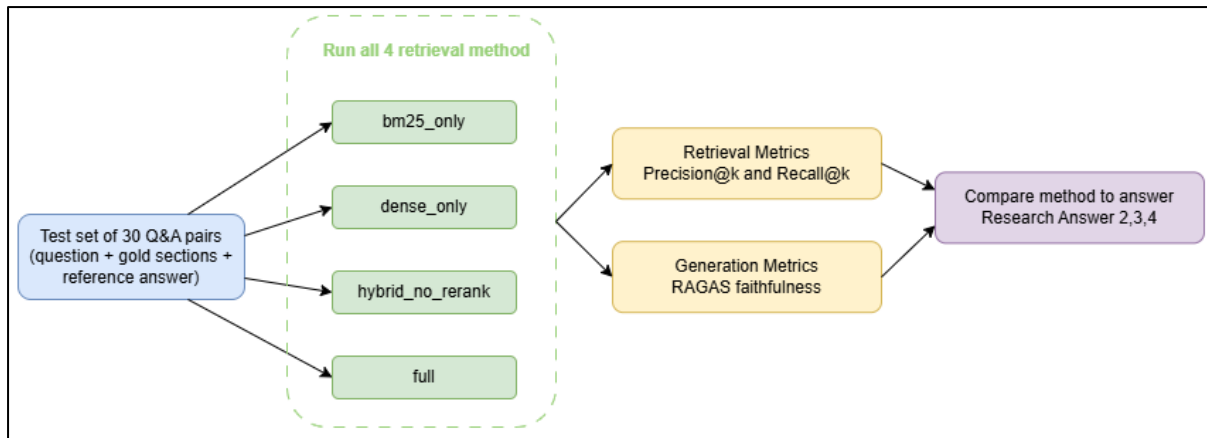


Figure 3. 6 Evaluation Design

3.7.1 Test Set Construction

Since there is currently lacking existing benchmark available for Malaysian statutory question answering, a test set will be created specifically for this project. It will consist of 30 pair of question and answer which obtained from the Employment Act 1955 and the Consumer Protection Act 1999. Each pair includes a question, a section or multiple section that used address the question, and a reference answer written based on those section(s). Four types of questions will be answered by the system to ensure its ability to handle different user information needs. Questions will be designed based on exact term including legal wording or section numbers, which sparse retrieval is expected to handle effectively on these types of questions. Paraphrased questions that are created in everyday language that is expected to handle best by semantic retrieval. Distractor questions that are prepared which keyword may also appear in the incorrect section to test the precision of the reranking stage, and out of scope questions to ensure that LexRAG-MY refuse to answer rather than producing hallucinate answer when no relevant section were found. These questions were spread across different parts of the documents within both Acts to avoid concentration on a single topic. Furthermore, some of the question include answer from single section while some question requires to be answered based on more than one sections to ensure that the metrics can reflect on different range of difficulty. The full

distribution of test set question table is shown in Table 3.4. However as mentioned in Section 1.5 as limitation, the test set will only include questions that have a single objectively correct answer to ensure the consistency of answers and to avoid legal bias. Lastly, each question answer pair will be check against the actual source of the Act to ensure that the identified section is genuinely answering the question before including it into the test set.

Type	Single section	Multiple section	Total
Exact term	6	2	8
Paraphrase	8	3	11
Distractor	5	2	7
Out of scope	4 (N/A section)		4
Total	23	7	30

Table 3. 4 Distribution of Question Type and Section Count

3.7.2 Retrieval Metrics (*Precision@k* and *Recall@k*)

The retrieval quality can be measured by Precision@k and Recall@k which compare the retrieved section with the gold section(s) of the test set. Precision@k measures the proportion of relevant section based on the question appears within the top-k results, with the formula, number of relevant sections in top-k divide by total number of sections retrieved. It captures how many irrelevant chunks are being passed to the LLMs. Recall@k, alternatively, measures the proportion of relevant items successfully retrieved within the top-k results, with the formula, number of relevant sections in top-k divide by total number of related sections. Precision@k capture whether the retrieval strategy reveals the section that the answer depends on. High recall value is expected for this system as the generated answer will depend on the retrieved sections that provides as context. Both metrics will be compared against multiple k value (k =1,3,5,10) to balance between retrieving insufficient information and avoiding unnecessary information being included. Furthermore, the metrics directly answering research question RQ2 to identify the best retrieval pipeline that locate the statutory question accurately based on plain language.

3.7.3 Generation Metrics (Faithfulness score using RAGAS)

Retrieval metrics alone do not tell whether the generated answer by LLMs is faithful to the retrieved chunks. Thus, the generation quality can be measured using RAGAS framework that was introduced in Section 2.6. It is an open-source framework that evaluate the faithfulness of RAG pipeline by decomposing each individual claims and compare it with the retrieved section. Then, it produces a faithfulness score range from 0 to 1 without requiring predefined answers written by humans. The RAGAS framework is well suited in this project as it evaluates exactly the faithfulness of the generated answer which the citation verification mechanism is designed to prevent unsupported statement from appearing in generated answers. The faithfulness score address research question RQ3 by comparing whether the generated answer is grounded in the retrieved context.

3.7.4 Baseline Comparison

The main part of the evaluation is to compare the performance of the four different retrieval strategies which answer research question R4 using the same test set, same generation stage and same verification component. Only the retrieval strategies are treated as independent variable, allowing the contribution of each retriever to be evaluated fairly. By comparing `bm25_only` and `dense_only` against `hybrid_no_rerank` reveals whether Naïve RAG can outperform advanced RAG in Malaysian statutory law settings. Additionally, comparing `hybrid_no_rerank` with the full proposed architecture demonstrate whether reranker can add value in retrieving the accurate chunks. While it is worth noting that reranker can reorder the retrieved chunks based on user question to improve the retrieve quality, the benefit of it should not automatically transfer in legal settings as highlighted in Section 2.5. The design of this evaluation ensures that only retrieval strategies are analysed without changing another component. This ensures that any observed improvement can be concluded based on result from the hybrid and reranking strategies rather than other unrelated factors. Table 3.5 summarizes all metrics that are expected to be collected for each retrieval strategy.

Retrieval Method	Retrieval stages used	Retrieval metrics	Generation metrics
<code>bm25_only</code>	BM25 only	Precision@k, Recall@k	RAGAS faithfulness

Retrieval Method	Retrieval stages used	Retrieval metrics	Generation metrics
dense_only	Dense only	Precision@k, Recall@k	RAGAS faithfulness
hybrid_no_rerank	BM25 + Dense + RRF	Precision@k, Recall@k	RAGAS faithfulness
full (proposed architecture)	BM25 + Dense + RRF + rerank	Precision@k, Recall@k	RAGAS faithfulness

Table 3. 5 Summary of the Four Retrieval Approach and the Evaluation Metrics

The citation verification mechanism itself is also expected to be evaluate as it acts as the final gate to determine the faithfulness of the generated answer. An unreliable judgement from verifier weakens the whole purpose of creating a reliable RAG system that is designed to reduce hallucinations. The LLM judge used in support check that discussed in Section 3.6 will be evaluated against 15 pairs of claims and cited statutory section and compares the LLM judge prediction with human label to match whether is it supported or not supported. The 15 pairs will be included both true and false cited section to ensure the LLM could capture it accurately. A confusion matrix would then be constructed to record the True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) value. Then, agreement rate will be calculated by dividing the pair which both judgements reach the same conclusion with the total claim and section pairs. The formula of agreement rate can be calculate using the equation:

$$Agreement\ Rate = \frac{TP + TN}{TP + TN + FP + FN}$$

This provides evidence on how the automatic process can be trusted when separate the performance measures with other component like retrieval and generation stages. The detailed evaluation structure that reviewed in this Chapter is shown in Appendix 1.

3.8 Tools and Technologies

The system will be implemented in Python due to its large number of libraries which comes with built-in modules for retrieval and evaluation that reduce the need to code from scratch, as well as its ability to integrate external APIs for the large language models. The implementation of system is prioritised with tools that run locally and free of charge, if possible, due to the cost constraint associated with student project. However, some of the cost like Claude API used for generation and judging purposes

is unavoidable. Each component and the technology used along with its reason for selection are summarized in Table 3.6.

Component	Technology Used	Reason for selection
Programming language	Python	Large choice of selection from libraries, and supporting the integration with external LLMs
PDF Extraction	PyMuPDF	Extract text from the 2 Acts that support local processing, remove unnecessary content that appear in the documents
Sparse retrieval	rank_bm25	Lightweight Python library that can perform sparse retrieval without external APIs.
Embeddings	Sentence-Transformers	Python library that converts text into numerical representation for semantic search
Vector database	Chroma DB	Vector database that able to store and retrieve embeddings that can be run locally without relying on external services
Reranking	BGE cross encoder reranker	Produce more accurate retrieval ranking by combining both chunks and question and score each of the pairs
Answer generation and citation	Claude API	Pay as you go pricing model that only charged based on usage. Use for constraint generation and citation.
Evaluation	RAGAS	Provide faithfulness score without reference answer
User Interface	Streamlit	Able to build simple chatbot demonstration using open-source Python framework
Version control	GitHub	Ensure that code is stored in repositories under version control, allowing previous code to be restored if necessary

Table 3. 6 Summary of Technology Stack

3.9 Ethical and Practical Considerations

Since the system provides legal information to Malaysian citizen, it is important to pay attention on ethical and practical considerations during the methodology phase rather than only implement it after fully development. LexRAG-MY is designed not to replace professional legal advice, but to provide citizen of Malaysia to gain legal information independently and to verify legal information that contain complex legal terminology from official statutory documents. A clear disclaimer on this effect is displayed in this report and will be present it on the interface of the system. Furthermore, the design of this proposed architecture that described in Section 3.5 ensures that the generated answer does not consider user's personal scenario which keep the safe boundary of only providing the information based on user query rather than providing legal advice.

In term of the data collection, both Malaysian statutory Act documents used are publicly available and were obtained from the Attorney General's Chambers portal, an official Malaysian government website that published the official copies of Malaysian

legal Acts. Moreover, LexRAG-MY does not collect any sensitive information from users because it is a one directional question answering system with no conversation history is stored, which was addressed as limitation on Section 1.5. The purpose of designing the system in this way is to ensure that user's questions could not be accumulated, which reduce the chances of LLM to hallucinate and provide extra assumptions cause by previous contexts.

Regarding the aspect of responsible Artificial Intelligence use, the designed architecture ensure that proposed system only provide answers based on retrieved chunks rather than hallucinate output. The failure handling function within the verification layer assure that user is never shown with a confident but an unfaithful answer. It is particularly crucial as mentioned in Section 1.6.3, which sensitive domain like law, medicine, engineering and other field require a reliable answer as errors made in these areas might leads to serious consequences. These considerations address the ethical, reliability and usability requirements that must be fulfil when developing a legal domain RAG system.

This chapter has explained how LexRAG-MY will be designed and evaluated. Section 3.2 present the Design Science Research approach and show the overall proposed architecture that is being divided into two architectures. The offline stage (Stage 0) where the two Acts will be indexed once and the online stage where each of the questions will be answered from retrieval to generation and verification components. Section 3.3 has explained how the two Malaysian Acts, Employment Act 1955 and the Consumer Protection Act 1999, will be obtained, extracted and divided into smaller chunks in the way that each of the chunks contains a complete statutory section with its own unique identifier that will be used for the verification steps.

Section 3.4 then fulfils the research objectives O2 by combining both sparse retrieval and dense retrieval through the Reciprocal Rank Fusion and enhanced the relevancy of top-k chunks based on user's question using a cross-encoder reranker as proposed retrieval architecture. Then, several simpler retrieval methods were introduced so that each retrieval performance can be measured and evaluated. Section 3.5 and Section 3.6 addressed objective O3 that describe how the answer will be generated and how each of the claims can be verified and traced back to the actual retrieved sections using a constrained prompt, with a single regeneration attempt and a failure

mechanism when verification fails. Section 3.7 explored the evaluation procedure which include the construction of test set and discussed on retrieval metrics like Precision@k and Recall@k, generation metric of RAGAS faithfulness and compare the full proposed pipeline with simpler baselines that addressed objective O4. Section 3.8 list down the tools that will be using to develop LexRAG-MY along with its reason of selection. Section 3.9 outlined the ethical and practical considerations including the system boundary, data handling method, and ensuring the reliability of AI generated answer to prevent hallucinations. These elements concluded the methodology which answer the research question Q2 to Q4. The work plan and timeline to implement this system will be further explored in Chapter 4.

Chapter 4: Work Plan and Timeline

4.1 Introduction

Chapter 3 have discussed on what will be built and how it will be evaluated. This chapter then discuss on how and when each part of the work will be carried out and arrange each of the scope that defined in Section 1.5 to be completed within the available time on Capstone Project 2. With a reliable schedule, the proposed system that described in Chapter 3 turn from an imaginary idea to a practical implementation plan.

This project will be conduct over a 13-week implementation timeline, and it will be breaks down into six phases based on the component outlined in Chapter 3. Phase A prepare the dataset for later stages of the project described in Section 3.3. Phase B develop the retrieval pipeline of Section 3.4 to address objective O2, while Phase C implements the constrained generation discussed in Section 3.5 and the verification method in Section 3.6 which fulfil objective O3. Phase D construct test set that specified in Section 3.7.1 for evaluation and Phase E perform testing on the reliability of the system and compare the proposed pipeline with the simpler RAG approach that defined in Section 3.7.2 to 3.7.4. Both phases deliver objective O4. Lastly, Phase F covers the demonstration interface and records the results and discussion.

In this chapter, Section 4.2 present the work breakdown structure that list down all the activities and the duration needed, define each of the dependencies and identify the critical path for this project. Section 4.3 discover the six milestones to provides checkpoints and track the progress of the project. Section 4.4 present the Gantt chart to show the activities, overlapping timeline, dependencies and the critical path on a single timeline. Section 4.5 identifies the possible risks when developing the system through a risk register and risk matrix.

4.2 Work Breakdown Structure (WBS)

Work Breakdown Structure (WBS) is a project management tool that used to divide a project into smaller and manageable tasks, also known as work packages, to ensure the project is easier to plan and track. The WBS decompose this project into 28 work packages that is grouped into six phases, as shown in Figure 4.1.

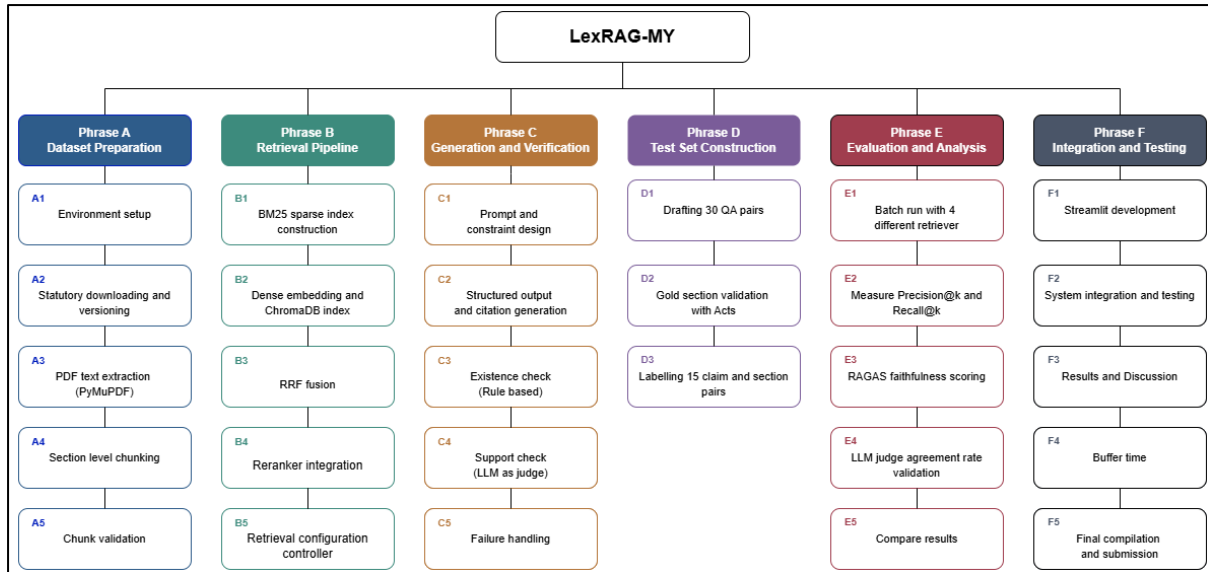


Figure 4. 1 Work Breakdown Structure for LexRAG-MY

The breakdown of the project is considered by two factors. First, the activities that do not depend on each other can be executed in parallel. For instance, the test set construction in Phase D and interface development in Phase F can be performed at the same time because creating the evaluation question does not require a chat interface to be completed. It is crucial especially when system need to be completed within a short period of developing time. Hence, both tasks can be scheduled concurrently to allocate sufficient time for evaluation phase. Second, some of the tasks that produce output which later will be used by many other tasks requires a separate checking steps to ensure that results are reliable. Hence, the chunk validation (A5) and the gold section validation (D2) are listed as separate verification tasks within WBS to prevent silent error that might cause unreliable evaluation results.

4.2.1 Lists of Activities

Table 4.1 identifies all the activities that is required to be completed within the 13-week implementation timeline, along with its unique identifier, duration of each activity and the justification behind the proposed duration. The period of each activity is decided based on the technical complexity that discussed in Chapter 3 rather than assigned each activity with a uniform timeline. For example, activities like section level chunking (A4) and test set drafting (D1) requires significant amount of time as compared with index construction for sparse and dense retrieval (B1 and B2).

ID	Activity	Duration	Reason of Duration
A1	Environment setup	0.5 week	Involving the installation of Python, and the libraries that are needed within Python environment, and setting up version control on GitHub. This is a technical preparation process, so half week is sufficient.
A2	Statutory downloading and versioning	0.5 week	Only two Acts were downloaded from AGC of Malaysia while records the exact publication date for each Act, ensuring the reproducibility of dataset as stated in Section 3.3. Both tasks can be done within the planned duration.
A3	PDF text extraction (PyMuPDF)	0.5 week	Most of the time allocated within the half week were mainly to inspect the extracted output, ensuring a clean and accurate text that will be carry into chunking stage. The extraction process itself is automated using the PyMuPDF Python library.
A4	Section level chunking	1.5 weeks	Longest activity in Phase A as it requires the chunking process to follow the Acts' section rather than performing fixed-length splitting. The rule-based parser must ensure all content were included within their identifier respectively across both Acts. Incorrect chunking approach could cause inaccurate results obtain in verification process.
A5	Chunk validation	0.5 week	Several sample of chunks will be manually compared against the actual sources to confirm the boundaries were accurate. It acts as quality check for milestone M1 that is listed in Section 4.3, before continuing to the next phase.
B1	BM25 sparse index Construction	0.5 week	Involving construction of BM25 index using rank_bm25, a lightweight Python library that require no implementing of algorithm from scratch.
B2	Dense embedding and ChromaDB index	1.0 week	Longer time needed as compared to B1 as it involves embedding of every chunk and store the vectors in ChromaDB. Section 3.4.2 require the embeddings to implement once during Stage 0 of architecture. Therefore, the correctness of generated embeddings is more important in order to return accurate chunks for dense retrieval.
B3	RRF fusion	0.5 week	RRF is simply a mathematical rule that combine retrieval that require no training data, as shown in Section 3.4.3. The duration allocated are mainly to construct the formula and to verify the ranked list produce by both retrievers.
B4	Reranker integration	1.0 week	The BGE cross encoder reranker will be loaded locally and produce relevance score for each question-chunk pair. Majority of time will be used for tuning like determining the candidate depth and the final top-k value
B5	Retrieval configuration controller	0.5 week	It acts as switch to select one of the four configuration parameters. It involves calling each retrieval function together, and it is necessary to answer research question RQ4 where

ID	Activity	Duration	Reason of Duration
			different retrieval method is compared while all other components remain the same.
C1	Prompt and constraint design	1.0 week	The design of system prompt must ensure that the generated answer is only based on retrieved context and refuse to answer if insufficient information is provided, using the RACE prompting framework. Additionally, constraint will be set such that the LLM must not provide any legal suggestion based on user personal scenario. The prompt will be tested and refined iteratively to ensure that it meets purpose of the system, which is why a full week of duration is allocated.
C2	Structured output and citation generation	1.0 week	Allowing the generated output follows a structured output that described in Section 3.5.2 with an error handling feature if LLM do not follow specific format before passing to verification component.
C3	Existence check (Rule based)	0.5 week	Rule based checking that compare the generated citation against the retrieved identifier that is described in Section 3.6. Low effort needed as it does not involve AI judgement.
C4	Support check (LLM as judge)	1.0 week	Longer implementation time than C3 as it involves judging mechanism that only evaluated based on claim made by generator and the actual content from the Acts, ensuring the faithfulness of the answer. Allocated duration also involved iterative prompting design until the judging behaviour is stable.
C5	Failure handling	0.5 week	Implement using stricter prompt if generated answer fails to pass through either existence or support check and assigning three types of status as mentioned in Section 3.6. Only single loop is expected rather than complex loop, so the development effort is small.
D1	Drafting 30 QA pairs	2.0 weeks	Each question must be able to be obtained from the two Acts, along with its gold section(s), and reference answer based on it. Thirty questions will be divided across four different question type that split between single and multiple section, as specified in Table 3.4. The activity has the largest duration allocation as compared to other activity and it does not depend on LexRAG-MY system implementation. Hence, it should work parallelly with other activity from Week 4, which will be illustrates in Gantt Chart in Section 4.4.
D2	Gold section validation with Acts	1.0 week	Since there is lack of existing benchmark available for Malaysian statutory question answering, each self-drafted pair will be manually check against the official Acts to ensure the section identified in gold section(s) genuinely answer the questions before includes it in the test set. The activity is separated from D1 and require a week because error in gold section would directly affect the evaluation metrics score silently.

ID	Activity	Duration	Reason of Duration
D3	Labelling 15 claim and section pairs	0.5 week	15 pairs of claims and section that consist of both accurate and inaccurate supported claims were drafted within half a week. This pairs will be used to evaluate accuracy of LLM used in support check and compute the agreement rate as described in Section 3.7.4.
E1	Batch run with 4 different retriever	1.0 week	Thirty questions will be processed through the 4 different retrieval approach while passing through using the same generation, verification and single retry path components. Time allocation includes the construction function for batch processing, ensuring all the question successfully completed before the evaluation process can starts.
E2	Measure Precision@k and Recall@k	0.5 week	The results returned from E1 will be evaluated using Precision@k and Recall@k with k value of 1,3,5 and 10 as specified in Section 3.7.2.
E3	RAGAS faithfulness scoring	1.0 week	RAGAS framework used to evaluate each generated answer by decomposing each individual claims and compare it with the retrieved section and returned a faithfulness score. The process takes longer than E2 as it requires model evaluation rather than rule-based comparisons.
E4	LLM judge agreement rate validation	0.5 week	Compares the dataset pairs produced in D3 against support check, construct a confusion matrix and computes the agreement rate. The duration allocate is less but essential because the credibility of verification component depends on it.
E5	Compare results	1.0 week	Interpret and compare the metrics across the four-retrieval strategy to draw conclusion on whether proposed architecture perform better as compare with simpler pipeline of RAG, addressing the research objective O4. One week is allocated because any anomalous result must be identified and solved before any conclusion drawn.
F1	Streamlit development	1.5 weeks	Implement chat interface as mentioned in Section 3.2 as user mode, fulfilling the main purpose of this system, which is providing a platform for layperson to access legal information. Furthermore, a clear disclaimer as discussed in Section 3.9 will be shown in the interface.
F2	System integration and testing	1.0 week	Final testing, ensuring system can switch smoothly between batch mode and user mode across four different retrieval pipelines. A week is allocated as error might occur when all the components are integrated.
F3	Results and Discussion	1.0 week	Record the results obtain and limitation as reference for future expansion or refine architecture that stated in Section 1.6.4.
F4	Buffer time	0.5 week	Extra time allocated for tasks with highly unpredictable completion times. The buffer time is placed at the end of project

ID	Activity	Duration	Reason of Duration
			rather than in the middle of project so that each activity duration remains honest.
F5	Final compilation and submission	0.5 week	Final checking of system and report including the formatting and appendix compilation before submission.

Table 4. 1 Activity Duration and Justification

The time allocated for each activity shows the difficulty of completing each task. Two of the activities allocated, which is section level chunking (A4) and drafting the question pairs (D1) consumed the most amount of time. Both activities are data preparation tasks rather than system implementation tasks. The reason of assigning huge effort on data preparation tasks was answered in Section 3.3 where the retrieval quality is depends heavily on sources quality, and because there is lack of Malaysian statutory benchmark existed as mentioned in Section 3.7.1. Hence, the process requires a manual construction and validation on the test set to ensure the reliability of metrics. However, several implementation activities like RRF fusion (B3) and existence check implementation (C3) requires only half a week to be completed since it is purely a mathematical ruled based construction rather than involving any training data process.

4.2.2 Activities Dependencies and Critical Path

Table 4.2 lists down all the activities with its dependencies and the reasons for it. This project implements a finish-to-start dependencies where the predecessor must be completed before the dependent activity begin. Both development environment setup (A1) and statutory downloading and versioning (A2) are not included in the table as both tasks do not require any prerequisites.

ID	Activity	Dependencies	Justification for the Dependencies
A3	PDF text extraction	A1, A2	Extraction cannot perform until the environment is set and documents are downloaded.
A4	Section level chunking	A3	Chunking will be operated on the extracted text from documents. Hence, extraction must be completed first.
A5	Chunk validation	A4	Validation process requires a complete set of chunks.
B1	BM25 sparse index Construction	A5	Both indexes were built using the same validated chunks, as mentioned in Section 3.3.
B2	Dense embedding and ChromaDB index	A5	

ID	Activity	Dependencies	Justification for the Dependencies
B3	RRF fusion	B1, B2	Fusion merge the output obtained from B1 and B2 and therefore both retrievers must be operational.
B4	Reranker integration	B3	Reranker is placed right after RRF which follow the retrieve then rerank approach that justified in Section 3.4.4.
B5	Retrieval configuration controller	B4	All retrieval method must exist before the switch mechanism can be developed and tested.
C1	Prompt and constraint design	B1	Prompt can only be designed once the retrieval output is available as the context and constraint must set strictly based on the retrieval output. B1 itself can supply sufficient information. Thus, this activity does not need to wait for the complete pipeline (B2, B3, B4 and B5) to be developed.
C2	Structured output and citation generation	C1	The structured of the output is set based on constraint prompt that developed in C1.
C3	Existence check (Rule based)	C2	The check compares the retrieved identifier with cited section produced by C2.
C4	Support check (LLM as judge)	C2	The check compares the retrieved section with each individual claim produced by C2.
C5	Failure handling	C3, C4	The retry is triggered by either check, so C3 and C4 must be implement first.
D1	Drafting 30 QA pairs	A5	Question drafted and gold section are written based on validated identifier, which the data must be reliable.
D2	Gold section validation with Acts	D1	Validation performs on the 30 question answer pairs.
D3	Labelling 15 claim and section pairs	C2, D2	Labelling of the claim and section pairs requires generated answers and its gold section(s).
E1	Batch run with 4 different retriever	B5	B5 provides a controller to change between different retrievers.
E2	Measure Precision@k and Recall@k	D2, E1	Output from E1 is compared against validated gold section from D2.
E3	RAGAS faithfulness scoring	E1	Faithfulness score is computed based on output produced by batch run.
E4	LLM judge agreement rate validation	D3, E1	Compare between LLM judge decision with human label.
E5	Compare results	E2, E3, E4	Comparison between retrievers require all metrics to be available.
F1	Streamlit development	C5	A complete system is expected before developing a chat interface.

ID	Activity	Dependencies	Justification for the Dependencies
F2	System integration and testing	F1	Integrating require both operating modes are completed.
F3	Results and Discussion	E5	Records the results and discuss based on different performance across all the metrics.
F4	Buffer time	F3	Place directly before submission to makes it easier to track whether earlier tasks have used extra time.
F5	Final compilation and submission	F2, F3	Submission includes both workable prototype and a detailed report writing.

Table 4. 2 Task Dependencies and Justification

Based on Table 4.2, the longest sequence of path that determine the earliest time this project can be completed, also known as the critical path, is A1/A2 → A3 → A4 → A5 → B1 → C1 → C2 → C3 & C4 → C5 → E1 → E3 → E5 → F3 → F4/F5. This indicates that activity that appear in the critical path must not be delay as it will directly delay the entire project completion date whereas delay in activity that does not occur in the path can be absorbed by slack.

This path is determined by project scheduling rather than technical complexity. For example, the core features like dense retrieval (B2), RRF fusion (B3) and reranker integration (B4) are not listed in the critical path as prompt design (C1) is depending on only construction of BM25 (B1) rather than waiting for the completion of pipeline at B5. Even if B2 to B5 completed earlier, the batch run process cannot be start as it still need to wait for the failure handling activity (C5). Therefore, C1 can start earlier to reduce the total project duration while allowing B2 to B5 can be delayed for 0.5 week.

The generation and verification phase (Phase C) are the critical since most of the activities appeared in the critical path. This information is useful as it decides when the project can be completed. Given this, extra time that obtained when some activities finish earlier than expected should be prioritized to prompt design, structured output and support check since delaying them might leads to postponing the overall project completion time. Conversely, delaying task like constructing reranker (B4) will not affect the overall progress as long as it finishes before the slack time. It also explains the reason of buffer time (F4) was placed before submission to ensure that the extra time will not be split among different activities. Thus, it ensures that the remaining time is consume only by activities appears in the critical path rather than simply use that on non-critical activities even when unnecessary.

4.3 Project Milestones

While activity consumed time, milestone acts as checkpoints on the timeline that mark a specific deliverable are completed. The six milestone define in Table 4.3 are positioned at the end of each project phase so that each deliverable is completed and validated before passing to the next phase where it depends on. Each milestone has its completion criteria that state a clear requirement for completion rather than estimating progress using an approximate percentage.

ID	Milestone	Week	Completion Criteria
M1	Validated statutory chunks	Week 4	Both Acts are chunked using the metadata schema that was presented in Table 3.1, and a set of samples are obtained to verified against the published Acts.
M2	Retrieval pipeline completion	Week 7	All four-retrieval method defined in Section 3.4 are constructed and returned a top-k ranked chunks, and allowing the methods to be change freely without altering any other components.
M3	Generation and verification integrated and test set finalized	Week 8	System produces a structured output that supported with section citation, both check can be execute with the single retry trigger function. All thirty-test set question is validation and available for evaluation.
M4	Completed evaluation	Week 11	All evaluation metric for the four configurations is collected and compile for comparison.
M5	Completed system integration	Week 12	The chat interface integrated for user mode while batch mode still enables for evaluation. Disclaimer that required by Section 3.9 is displayed.
M6	Final submission	Week 13	Final report that includes the results, discussion and appendices along with a workable prototype is submitted.

Table 4. 3 Project Milestone and Completion Criteria

The milestone M3 at Week 8 set as the most important checkpoint in this project. It requires both verification mechanism and a finalized test set to be completed simultaneously as the evaluation process as mentioned in Section 3.7 cannot begin until M3 is achieved. Hence, Phase C and Phase D will be scheduled on the same time despite both event serves different objectives. However, since Phase C lies on the critical path, it will cause the project delay if both phase face delay at the same time.

4.4 Gantt Chart

Figure 4.2 is a Gantt Chart that show the distribution of all the activities shown as horizontal bar across the 13 weeks' timeline. The Gantt Chart record the start week along with its duration for each activity and are categorized by their phases using different colours. The critical path for this project is marked and labelled in red thick border while the dependency that linked the critical activities are drawn in pink downward arrow. This allows the chain that identified the critical path in Section 4.2.2 can be followed across the timeline. Additionally, the diamonds that appear on the bottom of the timeline representing the milestones projected across the timeline.

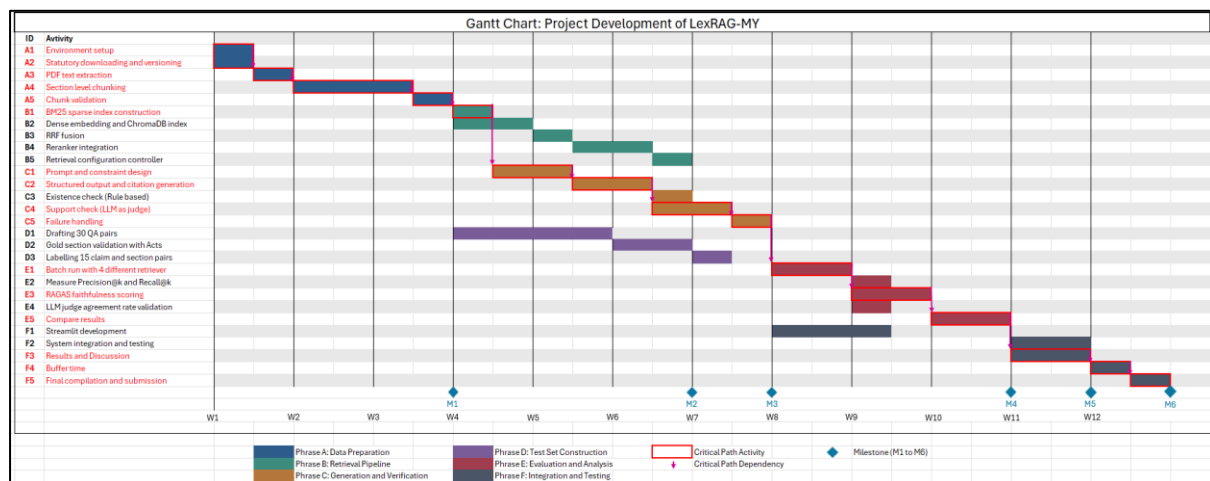


Figure 4. 2 Gantt Chart for LexRAG-MY

This chart reveals three characteristics of the project plan. First, some phases will be running parallelly. For example, Phase D starts in week 4 together with Phase B, and the Streamlit development (F1) begins in week 8 together with Phase E. This feature ensures that the entire work effort from data preparation to integration and testing can be completed within thirteen weeks. Second, the critical path shown in the Gantt chart reveals that it passes through Phase C and does not include activities from B2 to B5 in the critical path even though it contains important technical tasks such as the retrieval components, confirming the statement suggested in Section 4.2.2. Third, the buffer time (F4) was listed as part of the activity rather than omitted, so the project plan does not assume all the activity can be completed on time according to the schedule, allowing the possibility of unexpected delays without affecting the overall project completion duration.

4.5 Risk Considerations

While the duration discussed in Section 4.2 assumed that each activity proceeds as intended, this section reveal the possible risks that might cause overall project delay and the response strategies for each of the risks. The likelihood and the impact of risks will be accessed differently because risk with high likelihood but has low impact on the project would require a different mitigation strategy from one that have low likelihood but having high impact that prevents the system from meeting its objectives. Table 4.4 present the risk register with its likelihood, impact and the mitigation measure for each identified risk.

ID	Risk	Likelihood	Impact	Mitigation
R1	Inconsistent formatting of Acts like schedules and tables which prevents the accuracy of section-level chunking.	High	High	One and half weeks are allocated on A4 rather than only several days, along with a separate chunk validation process on A5 before proceeding to milestone M1, the earliest stage of milestone. Manual parsing for specific parts that fails is accepted rather delaying the entire process.
R2	The cross reranker does not improve the overall performance of RAG system based on Malaysian statutory text, as discussed in Section 2.5.3 as potential limitation.	Medium	Low	The retrieval architecture without reranker, hybrid_no_rerank, remain a valid configuration. It still answers the research question RQ4 rather than a project failure.
R3	API cost exceeded during evaluation with 30 test set of questions across four different retrieval approach when involve judge calls, RAGAS calls and possible retry when fails either of the checks.	Medium	Medium	Claude API uses pay as you go method for pricing model. A cost estimate will be produced before Phase E based on the expected input and output length for the LLM. Additionally, most of the component like sparse and dense indexing as implementation of the system is prioritized with tools that run locally.
R4	Test set construction requires longer time than expected as each pair must be verified	Medium	High	Start the construction process parallely with Phase B rather than waiting for the system to be completed. Separate activity of validation of the test set to ensure any incorrect pairs can be detected early.

ID	Risk	Likelihood	Impact	Mitigation
	against the actual source.			
R5	The LLM judge produces a weak human agreement rate which weakening the credibility of Objective O3.	Low	High	Iterative prompting design on C4 to ensure it truly understand the context and constraint of the system, and the agreement is measured at E4 using the 15 claim and section pairs before any conclusion on faithfulness of the system is drawn. If the agreement is still low, it will be declared as limitation of the system for future enhancement rather than remain concealed.
R6	Potential delay of project completion due to concurrent coursework from other subjects.	Medium	High	Construction of test set (Phase D) and Streamlit development (F1) that do not require any dependencies are conducted parallelized with other activities. Milestone are set as checkpoint to keep track of progress, and 0.5 week of buffer time (F4) is reserved as part of project timeline.

Table 4. 4 Risks Register

Plotting the likelihood and impact matrix on Figure 4.3 gives an illustration on the priority of each identified risk. Therefore, the risk with higher likelihood and higher impact should be flagged and prioritized to be solve as the project cannot be proceed until these risks are resolved.

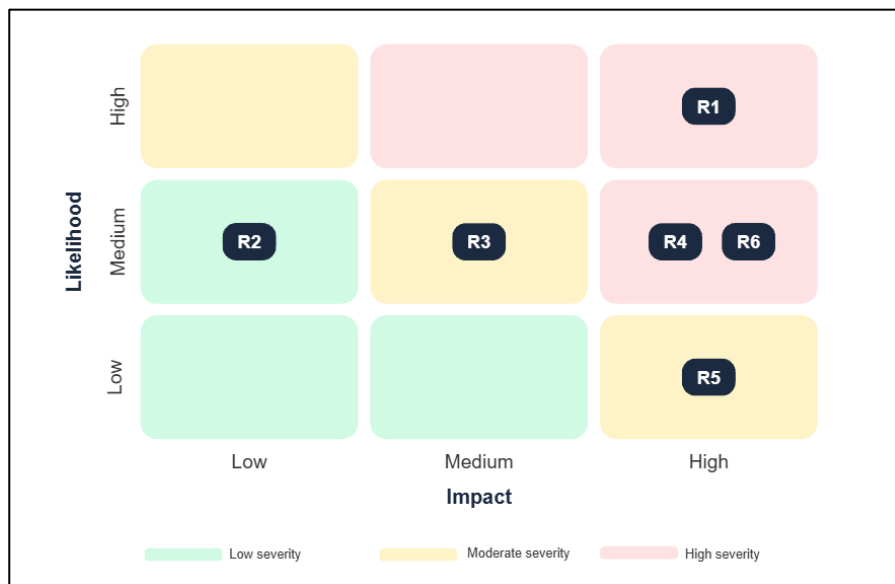


Figure 4. 3 Likelihood and Impact Matrix for LexRAG-MY

Two observations can be concluded based on the matrix. First, two out of three risks within the high severity zone, which is R1 and R4, are involved in data preparation rather than the system implementation. Both risks requiring human involvement on

statutory text, and the risks are harder to avoid as it depends on human review, with only R1 were flagged as high likelihood and high impact. Hence, the section level chunking and drafting of QA pairs was allocated for the longest duration in the schedule, with the validation process as separate tasks, and the buffer time are reserved to handle delays that might happen which caused by R1 and R4.

Second, R2 is the only one risk that sits alone in the low severity zone despite affecting the proposed architecture. This can be explained by Section 2.5 where it does not guarantee that reranking will help in Malaysia statutory legal setting and Table 4.4 described it as an optimization risk rather than project failure. The evaluation is designed to compare between proposed system with a simpler pipeline, which does not assure the effectiveness of the proposed system. So, an unfavourable result will still answer the research question RQ4.

This chapter converted the methodology in Chapter 3 into a realistic 13-week plan. Section 4.2 decomposed the project into 28 work packages across six phases from data preparation to integration and testing, which are shown in Figure 4.1, and list down the duration of each of the activities and the reason for that duration. That section also stated the finish-to-start dependencies of each activity with its justification. These dependencies then construct a critical path, showing the longest sequence of activities in the project with the shortest possible time needed to complete the project.

Section 4.3 defined the milestone with its completion criteria to ensure the progress can be tracked rather assumed. Section 4.4 presented a Gantt chart that list down all the activities, their overlaps timeline, the critical path and the proposed milestones in a single timeline. The Gantt chart exposed the data preparation steps (Phase D) and Streamlit development (F1) do not depend on any activity and therefore can begin parallelly with other tasks rather than arranging the timeline in sequence.

Section 4.5 identified all the possible risks that might encounter when implementing the project, assessed each on likelihood and impact, and the possible mitigation measure. Risks were presented in risk register and the likelihood and impact matrix, showing that two out of three risks that fall within the high severity zone involving manual data activities rather than system implementation, with R1 identified as the only risk with high likelihood and high impact. It is critical and action must be taken as the project cannot be proceed until these risks are resolved.

Collectively, Chapter 3 and Chapter 4 establishes the design of the project in detailed and the plan which it will be delivered on Capstone Project 2. The proposal is complete and the project can move forward to the implementation phase using the schedule that discussed in this chapter.

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Chapter 6: Appendices

Appendix 1.0 The Evaluation Structure

This appendix shows an evaluation structure that used to address Objective O4 that discussed in Section 3.7. It consists of six different table that records the 30 questions test set, retrieval scoring, generation scoring, out of scope checks, LLM judge validation and the summary of the results table.

Appendix 1.1 Test Set Structure

The test set includes 30 questions and answers pair which derived from both the Malaysian statutory Acts (Employment Act 1955 and Consumer Protection Act 1999), with 8 Exact Term, 11 Paraphrase, 7 Distractor and 4 Out of scope type of questions. Total of 30 rows with 8 columns are expected to be fill.

QID	Question (Plain Language)	Question Type	Section Type	Act	Gold Section(s)	Gold Count	Reference Answer
Q01	question	Exact Term	Single	Employment Act 1955	60A(1)	1	answer
Q02	question	Exact Term	Multiple	Consumer Act Protection	24S(1); 24U(1)	2	answer
⋮							
Q30	question	Out of Scope	N/A	N/A	0	0	answer

Figure A1. 1 Sample Structure of the Evaluation Test Set

Appendix 1.2 Retrieval Score

Retrieval scores involve both Precision@k and Recall@k where k = 1,3,5,10 for all 26 in-scope questions and record each result across four different retrieval pipelines (bm25_only, dense_only, hybrid_no_rerank and full). Total of 104 rows with 11 columns are expected to be fill.

QID	Retrieval	Gold Count	P@1	P@3	P@5	P@10	R@1	R@3	R@5	R@10
Q01	bm25_only	1								
Q01	dense_only	1								
Q01	hybrid_no_rerank	1								
Q01	full	1								
⋮										
Q26	full	2								

Figure A1. 2 Retrieval Evaluation Template

Appendix 1.3 Generation Score

Generation scoring records the RAGAS faithfulness score with all the in-scope questions across all four different retrieval pipelines along with its verification status. Total of 104 rows with 8 columns are expected to be fill.

QID	Retrieval	Faithfulness (0-1)	Claims Total	Claims Pass	Verification Status	Retry Triggered?	Answer Shown?
Q01	bm25_only		1	1	verified	N	Y
Q01	dense_only		1	1	verified_after_retry	Y	Y
Q01	hybrid_no_rerank		1	1	verified	N	Y
Q01	full		0	0	insufficient_info	Y	N
:							
Q26	full		1	1	verified_after_retry	Y	Y

Figure A1. 3 Generation and Citation Verification Template

Appendix 1.4 Out of Scope Refusal Decision

The 4 out of scope questions (Q27 to Q30) will be evaluated separately as the answer could not be found in the 2 Acts. The system is expected to refuse to answer by answering “Insufficient information in the Acts” rather than producing a hallucinate answer with a retrieval scoring. Total of 16 rows with 4 columns are expected to be fill.

QID	Retrieval	System Output	Correct Refuse Output?
Q27	bm25_only	Insufficient Information in the Acts	Yes
Q27	dense_only	Insufficient Information in the Acts	Yes
Q27	hybrid_no_rerank	Insufficient Information in the Acts	Yes
Q27	full	Insufficient Information in the Acts	Yes
:			
Q30	full		No

Figure A1. 4 Out of Scope Evaluation Template

Appendix 1.5 LLM Judge Validation

The LLM judge used in support check within the citation verification mechanism will be validate separately against 15 claim text and cited section pair, showing the reliability of the system. Total of 15 rows with 7 columns are expected to be fill.

Pair	Claim Text (by Generator)	Cited Section(s)	Cited Section Text	Human Label	Judge Label	Agree?
P1				Not supported	Supported	No
P2				Supported	Not supported	No
P3				Supported	Supported	Yes
:						
P15				Not supported	Not supported	Yes

Figure A1. 5 LLM Judge Validation Template

A confusion matrix will then create based on the output above, and agreement rate will be calculated by dividing the TP + TN with TP + TN + FP + FN.

	Judge: Supported	Judge: Not supported
Human: Supported	TP value	FN value
Human: Not supported	FP value	TN value
Agreement Rate:		50%

Appendix 1.6 Summary Table

Results obtained from Appendix 1.1 to 1.5 will be summarized into a single table that used to compare different retrieval strategies with proposed architecture and answer the research question RQ2 to RQ4. Total of 4 rows with 13 columns are expected to be fill.

Retrieval	Retrieval Metrics								Generation Metrics		Out of Scope		LLM Judge
	P@1	P@3	P@5	P@10	R@1	R@3	R@5	R@10	Mean Faithfulness	Verified Rate	Refuse Output	Accuracy	Agreement rate
bm25_only													
dense_only													
hybrid_no_rerank													
full													

Figure A1. 6 Summary Comparison of Retrieval Approaches Across All Evaluation Metrics